

Sparkling Clean

University Cleaning Inventory & Issuance System PRG381 Group Project, Group 3

Sparkling Clean is our Track B submission for the PRG381 Project.

It's a Java Swing desktop app that lets university staff manage cleaning materials, suppliers, cleaners, and stock issuances. Built to meet the assignment's required modules (authentication, dashboard, CRUD, stock issuance, reports, and role-based access).

Table of Contents

- [Tech Stack](#)
 - [Project Structure](#)
 - [Package Breakdown](#)
 - [How the UI fits together](#)
 - [Role-Based Access Control](#)
 - [How role-based access is implemented](#)
 - [Owner](#)
 - [Storekeeper](#)
 - [Cleaner](#)
 - [Login and Register Screenshots](#)
 - [Database Schema \(ERD\)](#)
 - [Security](#)
 - [Getting Started](#)
 - [Prerequisites](#)
 - [Installation](#)
 - [Database Setup](#)
 - [Opening the Project in NetBeans](#)
 - [Running the project](#)
 - [Resolving Errors](#)
 - [1. Adding the PostgreSQL JDBC Driver \(By default it should be there\)](#)
 - [2. Fixing JDK Version Errors in NetBeans](#)
 - [3. Troubleshooting Common Setup & Runtime Issues](#)
 - [Team, Group 3](#)
 - [Contributors & Code Attribution](#)
-

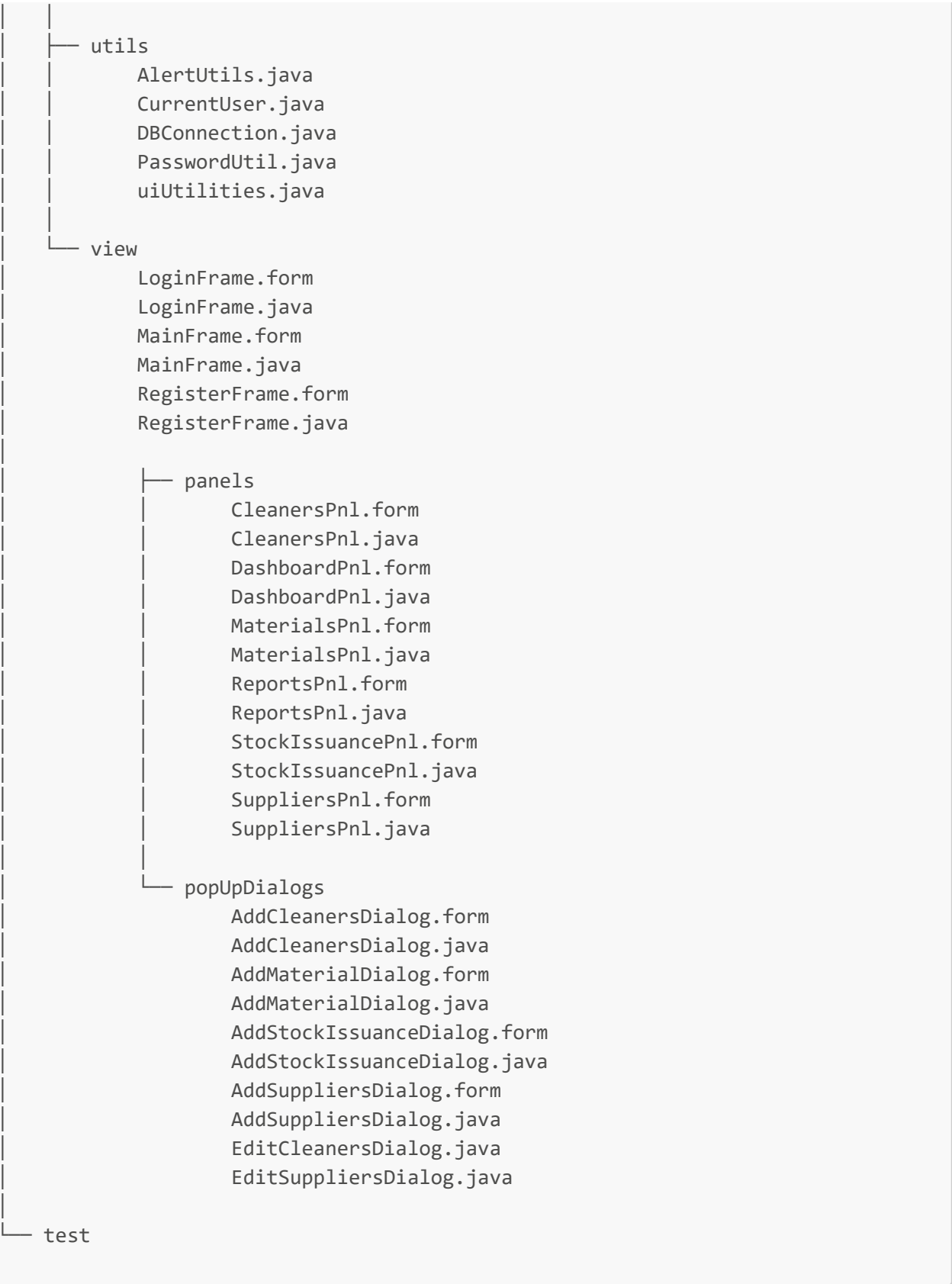
Tech Stack

Layer	Technology
Language	Java
GUI	Java Swing (built with NetBeans GUI Builder)
Data Access	JDBC

Layer	Technology
Database	PostgreSQL 18
IDE	Apache NetBeans 30
JDK	JDK 26
Operating System	Windows (developed and tested on Windows)

Project Structure

```
/PRG381_Project_PTA381_AM_Group-3
| .gitignore
| PRG381_Project_PTA381_AM_Group-3.iml
| README.md
|— .idea
|— images
|— University Cleaning Inventory And Issuance System
|   |— build
|   |— database
|   |   | ERD.jpeg
|   |   | schema_with_mockdata.sql
|   |— dist
|   |— lib
|   |— nbproject
|   |— src
|   |   |— controller
|   |   |   | AuthController.java
|   |   |   | CleanerController.java
|   |   |   | MaterialController.java
|   |   |   | MaterialDAOInterface.java
|   |   |   | StockIssuanceController.java
|   |   |   | SupplierController.java
|   |   |— dao
|   |   |   | CleanerDAO.java
|   |   |   | ReportsDAO.java
|   |   |   | StockIssuanceDAO.java
|   |   |   | SupplierDAO.java
|   |   |   | UserDAO.java
|   |   |— model
|   |   |   | Cleaner.java
|   |   |   | Material.java
|   |   |   | StockIssuance.java
|   |   |   | Supplier.java
|   |   |   | User.java
|   |   |— services
|   |   |   | CleanerService.java
|   |   |   | SupplierService.java
```



Package Breakdown

Package / Folder	Purpose
view	Top-level Swing frames (LoginFrame , RegisterFrame , MainFrame) that manage application startup, authentication screens, and the main layout shell with sidebar navigation.

Package / Folder	Purpose
<code>view.panels</code>	Embedded <code>JPanel</code> components swapped inside <code>MainFrame</code> 's main layout (<code>DashboardPnl</code> , <code>MaterialsPnl</code> , <code>SuppliersPnl</code> , <code>CleanersPnl</code> , <code>StockIssuancePnl</code> , <code>ReportsPnl</code>). Each panel presents the actual screen for that page.
<code>view.popUpDialogs</code>	Modal GUI dialogs (<code>JDialog</code>) used for creating and editing records (<code>AddCleanersDialog</code> , <code>EditSuppliersDialog</code> , <code>AddStockIssuanceDialog</code> , etc.) without navigating away from the active panel.
<code>controller</code>	Application controllers containing business logic and intermediary validation between the presentation views (<code>view</code>) and data persistence layer (<code>dao</code>).
<code>dao</code>	Data Access Object layer (<code>CleanerDAO</code> , <code>SupplierDAO</code> , <code>ReportsDAO</code> , <code>UserDAO</code> , etc.) encapsulating raw SQL queries and database CRUD operations.
<code>model</code>	Model classes (<code>Cleaner</code> , <code>Material</code> , <code>Supplier</code> , <code>StockIssuance</code> , <code>User</code>) mapping database tables to structured Java objects.
<code>services</code>	Service layer components handling higher-level business rules and orchestration across DAO operations (e.g., <code>CleanerService</code> , <code>SupplierService</code>).
<code>utils</code>	Shared helper utilities: <code>DBConnection</code> for centralized JDBC connectivity, <code>PasswordUtil</code> for secure hashing, <code>CurrentUser</code> for session and role state tracking, and <code>AlertUtils</code> / <code>uiUtilities</code> for UI styling and notifications.
<code>database</code>	Storage for system database artifacts, including the ER Diagram (<code>ERD.jpeg</code>) and initialization scripts (<code>schema_with_mockdata.sql</code>).
<code>nbproject</code>	Auto-generated NetBeans project configuration and build metadata.

How the UI fits together

The application is built around three top-level `JFrame` components in `ui/`:

1. `LoginFrame`, the application's entry point (`main.class`). Handles staff login and links to registration.
2. `RegisterFrame`, handles new staff member registration.
3. `MainFrame`, loaded after a successful login. This is the main application window.

`MainFrame` is split into two areas:

- A left-hand sidebar with the logo, navigation buttons (Dashboard, Materials, Suppliers, Cleaners, Stock Issuance, Reports), and logged-in user info.
- A right-hand content area (`contentPanel`), a `CardLayout` container that swaps between the six `JPanel` forms in `ui/panels/`, depending on which sidebar button is clicked.

Each panel in `ui/panels/` is a self-contained `JPanel` responsible for one module's CRUD screen and logic (e.g. `MaterialsPnl` handles adding, editing, and deleting cleaning materials), which keeps `MainFrame` itself focused purely on layout and navigation.

Role-Based Access Control

There are three roles: Owner, Storekeeper, and Cleaner.

Which sidebar buttons appear and which page actions are enabled depend entirely on the role of the currently logged-in staff member.

The mock data (loaded via the SQL script in [Database Setup](#)) includes a ready-to-use login for each role, see the credentials table in the [Running the project](#) section to try each role out.

How role-based access is implemented

`src/utils/CurrentUser.java` keeps track of whoever is currently logged in and offers simple yes/no checks for their role:

```
public class CurrentUser {
    public static final String ROLE_CLEANER = "Cleaner";
    public static final String ROLE_STOREKEEPER = "Storekeeper";
    public static final String ROLE_OWNER = "Owner";
    private static User loggedInUser;

    private CurrentUser() {
    }

    public static void set(User user) {
        loggedInUser = user;
    }

    public static void clear() {
        loggedInUser = null;
    }

    public static User get() {
        return loggedInUser;
    }

    public static boolean isLoggedIn() {
        return loggedInUser != null;
    }

    public static boolean hasRole(String role) {
        return loggedInUser != null &&
        loggedInUser.getRole().equalsIgnoreCase(role);
    }

    public static boolean isCleaner() {
        return hasRole(ROLE_CLEANER);
    }

    public static boolean isStorekeeper() {
        return hasRole(ROLE_STOREKEEPER);
    }
}
```

```

    public static boolean isOwner() {
        return hasRole(ROLE_OWNER);
    }
}

```

Each panel uses these checks to decide what to show. There are two patterns depending on the page:

1. **Dashboard and Reports**, Owner and Storekeeper see exactly the same thing here (full access, no restrictions), so these two panels don't need any role checks at all. The Cleaner never sees these pages because the sidebar itself doesn't show Dashboard or Reports buttons for that role, so there's nothing to restrict once you're on them.
2. **Materials, Suppliers, Cleaners, Stock Issuance**, access differs between roles on these pages, so each of these panels calls its own `applyRoleRestrictions()` method right after `initComponents()` in its constructor. This method checks the role and then hides or disables whatever that role isn't allowed to use. For example, in `CleanersPnl`, only the Storekeeper can add, edit, or delete, so the Owner gets the "Add Cleaner" button hidden and the Edit/Delete table columns removed entirely:

```

public CleanersPnl() {
    initComponents();
    applyRoleRestrictions();
}

private void applyRoleRestrictions() {
    boolean canEdit = CurrentUser.isStorekeeper();

    btnAddCleaner.setEnabled(canEdit);
    btnAddCleaner.setVisible(canEdit);

    if (!canEdit) {
        javax.swing.table.TableColumnModel columnModel = jTable1.getColumnModel();
        columnModel.removeColumn(jTable1.getColumnModel().getColumn("Edit"));
        columnModel.removeColumn(jTable1.getColumnModel().getColumn("Delete"));
    }
}

```

Every other CRUD panel follows this same type of pattern. A role never sees a control it isn't allowed to use: buttons are disabled and set to not visible, and columns are removed outright, not just greyed out.

The screenshots under each role below show what these restrictions look like side by side, compare the same page across roles to see which buttons/columns disappear.

Owner

Starting screen: **A blank pannel, user must choose a option to navigate to**

Module	Access
Dashboard	Full view

Module	Access
Materials	View only
Suppliers	Create, Read, Update, Delete
Cleaners	View only
Stock Issuance	View only
Reports	Full access

The Owner's sidebar shows all modules. Dashboard and Reports are identical to what the Storekeeper sees (no restrictions on either role there); the difference between the two roles shows up on Materials, Suppliers, Cleaners, and Stock Issuance, where the Owner is read-only everywhere except Suppliers.

Dashboard

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

admin

Owner

Log out

Dashboard

Overview of your cleaning supplies inventory

Inventory Value

R 3,159.00

Across 8 material types

Total Materials

8

of 190 units total

Low Stock Items

4

Needs attention

Active Cleaners

6

1 inactive

Recent Issuances

0

in the Last 7 days

Low stock items

Material	Current	Min	Supplier
Hand Sanitizer	3	10	EcoClean Solutions
Glass Cleaner	5	10	EcoClean Solutions
Microfiber Cloths	8	10	Janitorial Essentials
Floor Disinfectant	12	15	CleanPro Supplies

Recent Stock Issuances

Material	Issued To	Quantity	Date
All-Purpose Cleaner	Maria Santos	5	2026-06-28 09:15
Microfiber Cloths	James Wilson	2	2026-06-27 11:30
Disposable Gloves	Patricia Garcia	3	2026-06-26 14:00
Floor Disinfectant	Robert Johnson	4	2026-06-25 10:45
Trash Bags (Large)	Maria Santos	3	2026-06-24 08:30
Glass Cleaner	James Wilson	2	2026-06-23 13:10
Mop Heads	Patricia Garcia	4	2026-06-22 15:20

Materials (View Only)

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

admin

Owner

Log out

Materials

Manage your cleaning materials inventory

Search materials ...

Search

All Categories

Total Items

8

Low Stock

4

Total Units

190

Material	Catergory	Quantity	Reorder Level	Supplier	Unit Cost
All-Purpose Cleaner	Cleaners	45	20	CleanPro Supplies	R 8.50
Floor Disinfectant	Disinfectants	12	15	CleanPro Supplies	R 12.00
Microfiber Cloths	Tools	8	10	Janitorial Essentials	R 25.00
Disposable Gloves	Safety	35	15	Janitorial Essentials	R 18.00
Glass Cleaner	Cleaners	5	10	EcoClean Solutions	R 9.50
Mop Heads	Tools	22	8	Janitorial Essentials	R 15.00
Trash Bags (Large)	Consumables	60	20	CleanPro Supplies	R 22.00
Hand Sanitizer	Hygiene	3	10	EcoClean Solutions	R 35.00

Suppliers (Full CRUD)

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

admin

Owner

Log out

Suppliers

Manage your cleaning supply vendors

Search suppliers by name, contact, or email ...

Search

Add new supplier

Total Suppliers

3

Active

3

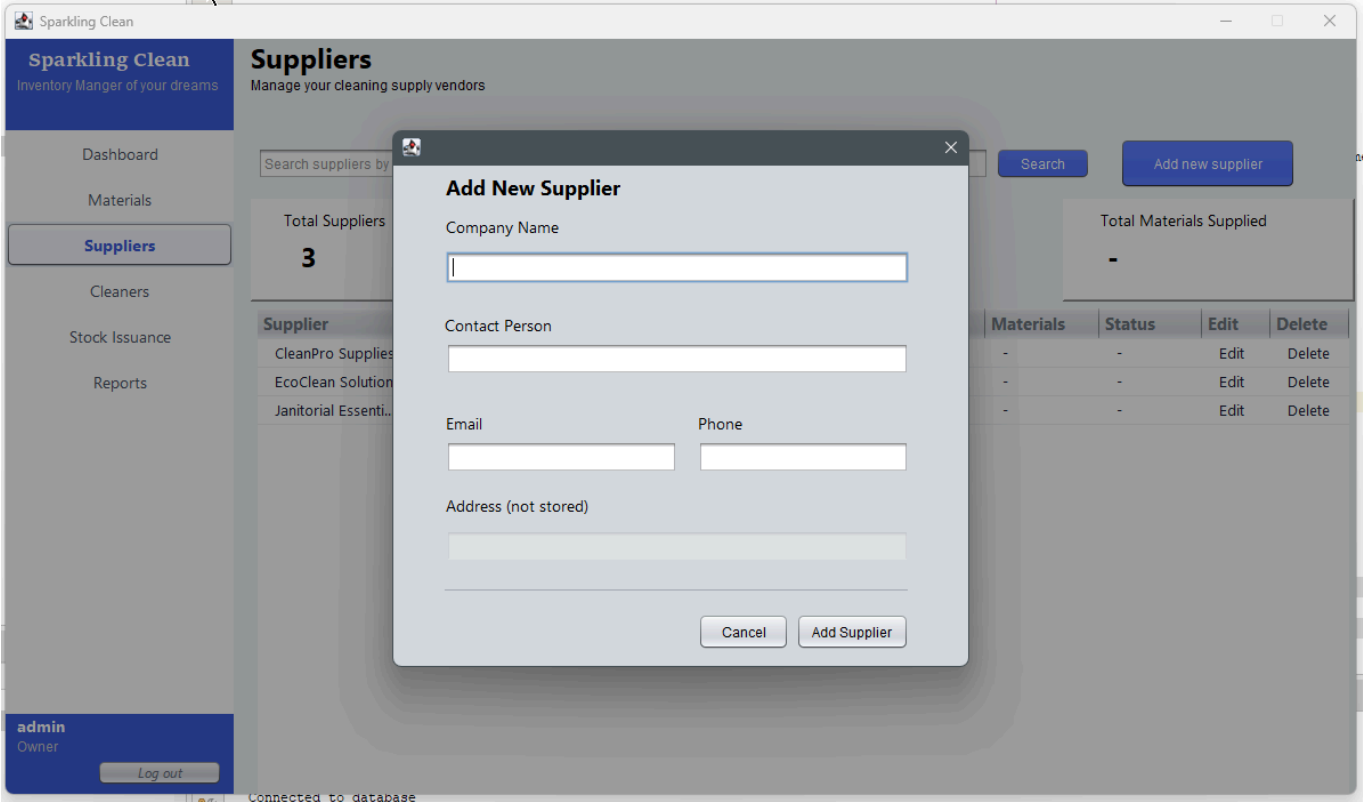
Total Materials Supplied

-

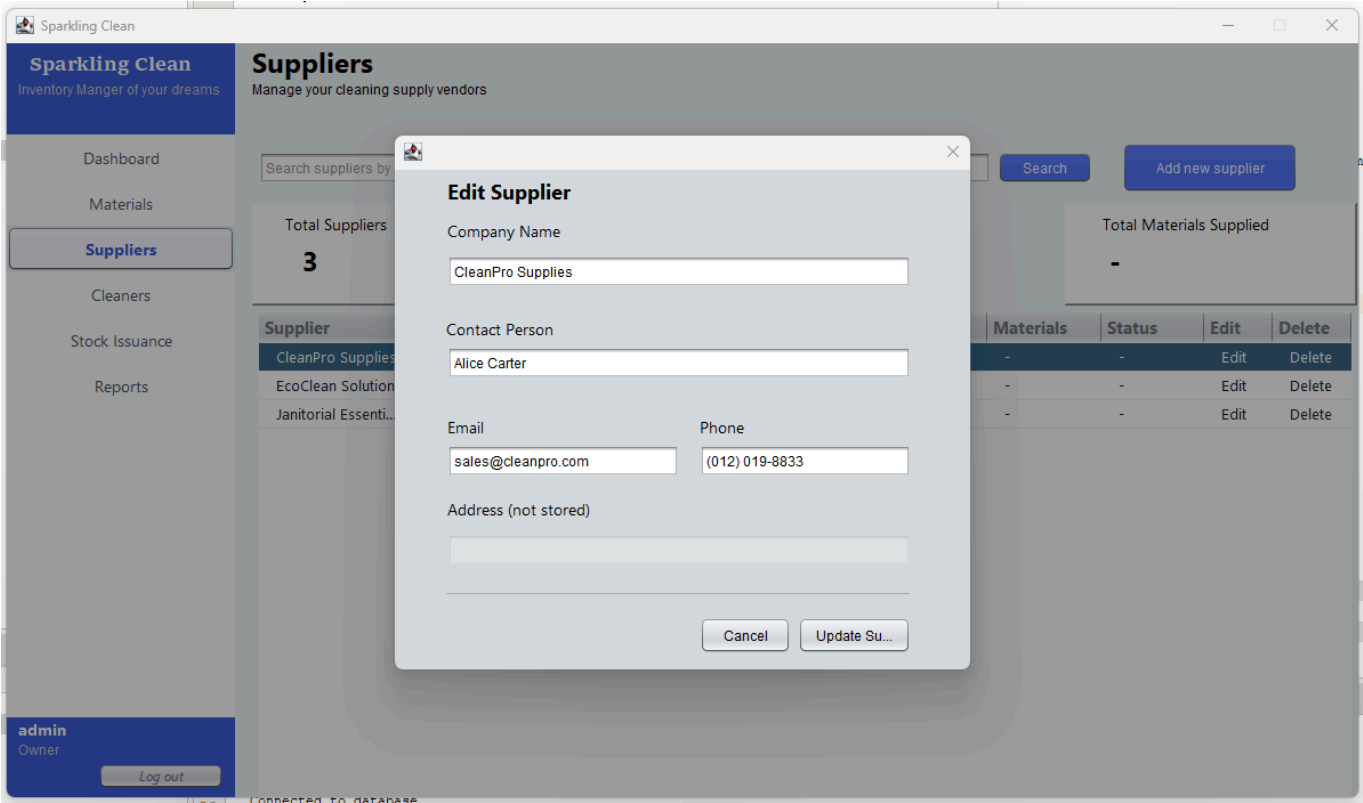
Supplier	Email	Phone number	Contact Person	Materials	Status	Edit	Delete
CleanPro Supplies	sales@cleanpro.com	(012) 019-8833	Alice Carter	-	-	Edit	Delete
EcoClean Solutions	support@ecocleansolutio...	(012) 017-7744	Diana Prince	-	-	Edit	Delete
Janitorial Essenti...	orders@janitorialessentia...	(012) 014-9922	Bob Miller	-	-	Edit	Delete

Supplier Pop-ups

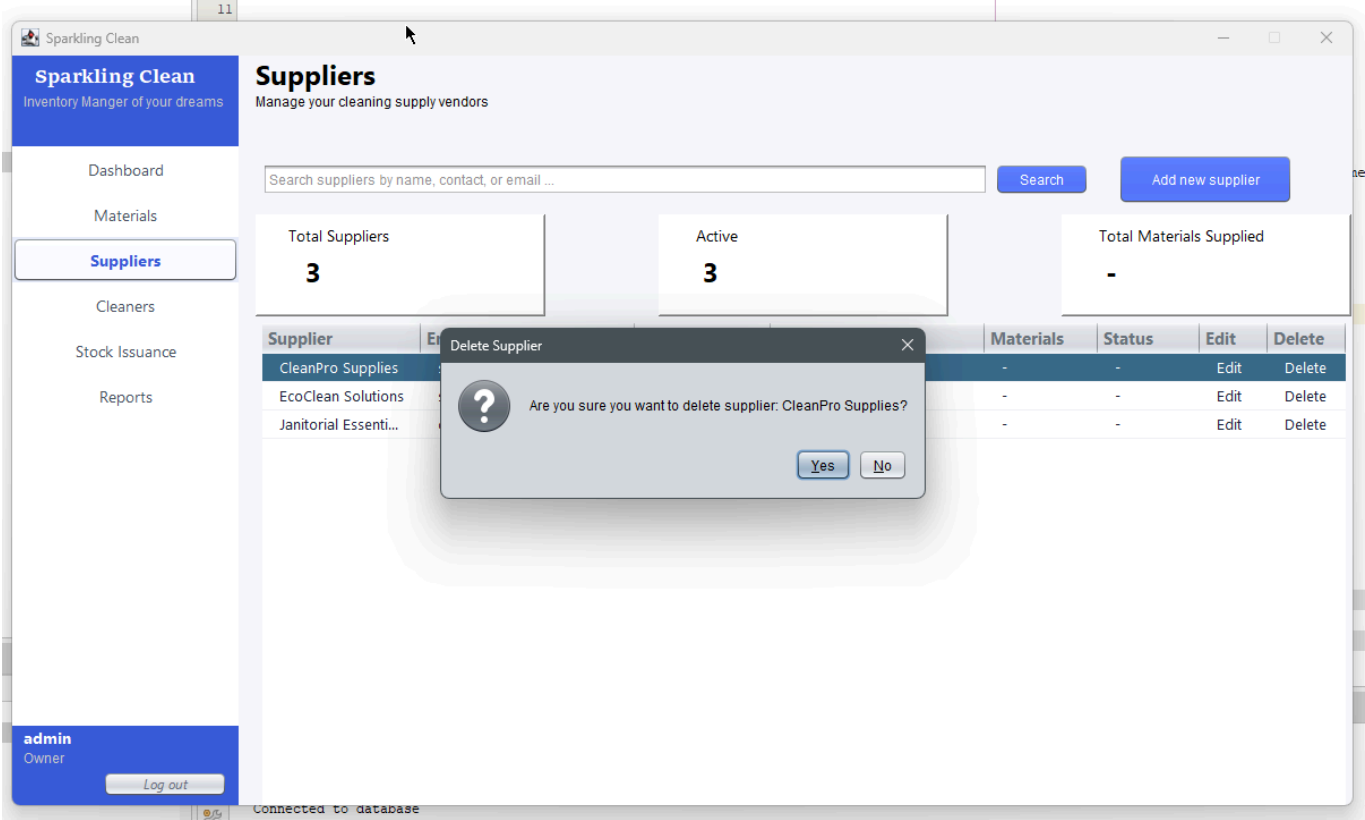
Add Supplier (Create)



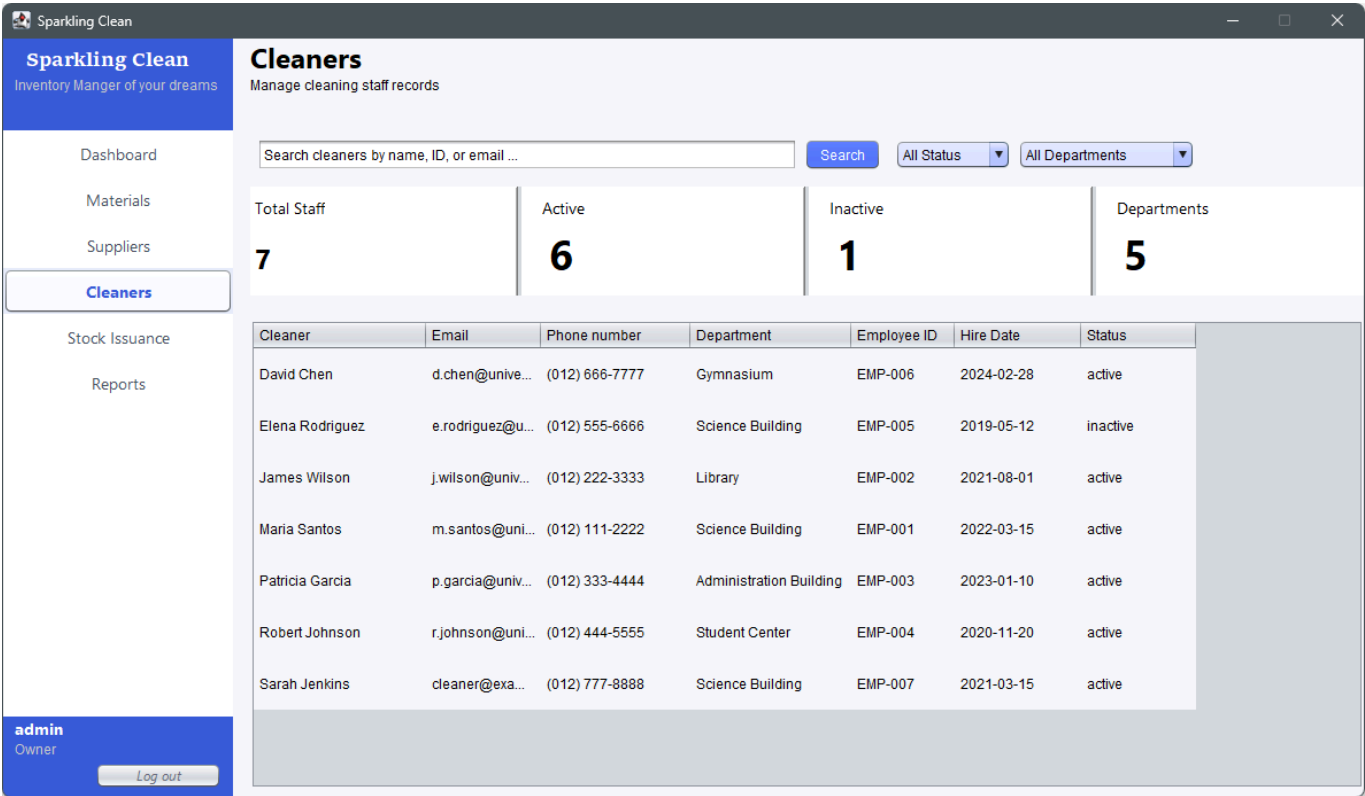
Edit Supplier (Update)



Delete Supplier (Delete)



Cleaners (View Only)



Stock Issuance (View Only)

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

admin

Owner

Log out

Stock Issuance

Check out the needed cleaning materials

Search by material ...

Search

Total Issuances

22

Total Items Issued

55

Low Stock items

4

Date	Material	Issued To	Quantity	Issued By	Notes
2026-06-28 09:15:00.0	All-Purpose Cleaner	Maria Santos	5	jdoe	Weekly su...
2026-06-27 11:30:00.0	Microfiber Cloths	James Wilson	2	jdoe	Replace w...
2026-06-26 14:00:00.0	Disposable Gloves	Patricia Garcia	3	jdoe	Routine re...
2026-06-25 10:45:00.0	Floor Disinfectant	Robert Johnson	4	jdoe	Deep clea...
2026-06-24 08:30:00.0	Trash Bags (Large)	Maria Santos	3	jdoe	Regular st...
2026-06-23 13:10:00.0	Glass Cleaner	James Wilson	2	jdoe	Library wi...
2026-06-22 15:20:00.0	Mop Heads	Patricia Garcia	4	jdoe	Administr...
2026-06-21 09:05:00.0	Hand Sanitizer	Robert Johnson	2	jdoe	Hand sani...
2026-06-20 10:00:00.0	Mop Heads	Elena Rodriguez	2	jdoe	Mop hea...
2026-06-19 14:30:00.0	All-Purpose Cleaner	Sarah Jenkins	3	jdoe	Weekly su...
2026-06-18 08:45:00.0	Floor Disinfectant	Maria Santos	2	sjenkins	Floor disi...
2026-06-17 11:15:00.0	Trash Bags (Large)	James Wilson	2	sjenkins	Trash bag...
2026-06-16 13:40:00.0	All-Purpose Cleaner	Patricia Garcia	4	jdoe	General cl...
2026-06-15 09:50:00.0	Microfiber Cloths	Robert Johnson	1	jdoe	Microfibe...
2026-06-14 10:20:00.0	Disposable Gloves	Elena Rodriguez	2	jdoe	Gloves for...
2026-06-13 15:00:00.0	Glass Cleaner	Sarah Jenkins	1	sjenkins	Glass clea...

Reports (Full Access)

Inventory Report

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

admin

Owner

Log out

Reports

View different inventory reports

Export to CSV

Inventory Report

Low Stock Report

Issuance History

Material Usage

Total Materials

8

Total Units

190

Total Value

R3,159.00

Low Stock Items

4

Search material or supplier

Search

Total Value ASC ↑

Material	Category	Quantity	Reorder Level	Supplier	Unit Cost	Total Value	Status
All-Purpose Cleaner	Cleaners	45	20	CleanPro Supplies	R 8.50	R 382.50	OK
Disposable Gloves	Safety	35	15	Janitorial Essentials	R 18.00	R 630.00	OK
Floor Disinfectant	Disinfectants	12	15	CleanPro Supplies	R 12.00	R 144.00	Low Stock
Glass Cleaner	Cleaners	5	10	EcoClean Solutions	R 9.50	R 47.50	Low Stock
Hand Sanitizer	Hygiene	3	10	EcoClean Solutions	R 35.00	R 105.00	Low Stock
Microfiber Cloths	Tools	8	10	Janitorial Essentials	R 25.00	R 200.00	Low Stock
Mop Heads	Tools	22	8	Janitorial Essentials	R 15.00	R 330.00	OK
Trash Bags (Large)	Consumables	60	20	CleanPro Supplies	R 22.00	R 1,320.00	OK

Low Stock Report

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

admin

Owner

Log out

Reports

View different inventory reports

Export to CSV

Inventory ReportLow Stock ReportIssuance HistoryMaterial Usage

4 items below reorder level

Immediate attention required for restocking

Material	Category	Current	Reorder Level	Shortage	Supplier
Hand Sanitizer	Hygiene	3	10	7	EcoClean Solutions
Glass Cleaner	Cleaners	5	10	5	EcoClean Solutions
Floor Disinfectant	Disinfectants	12	15	3	CleanPro Supplies
Microfiber Cloths	Tools	8	10	2	Janitorial Essentials

Issuance Report

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

admin

Owner

Log out

Reports

View different inventory reports

Export to CSV

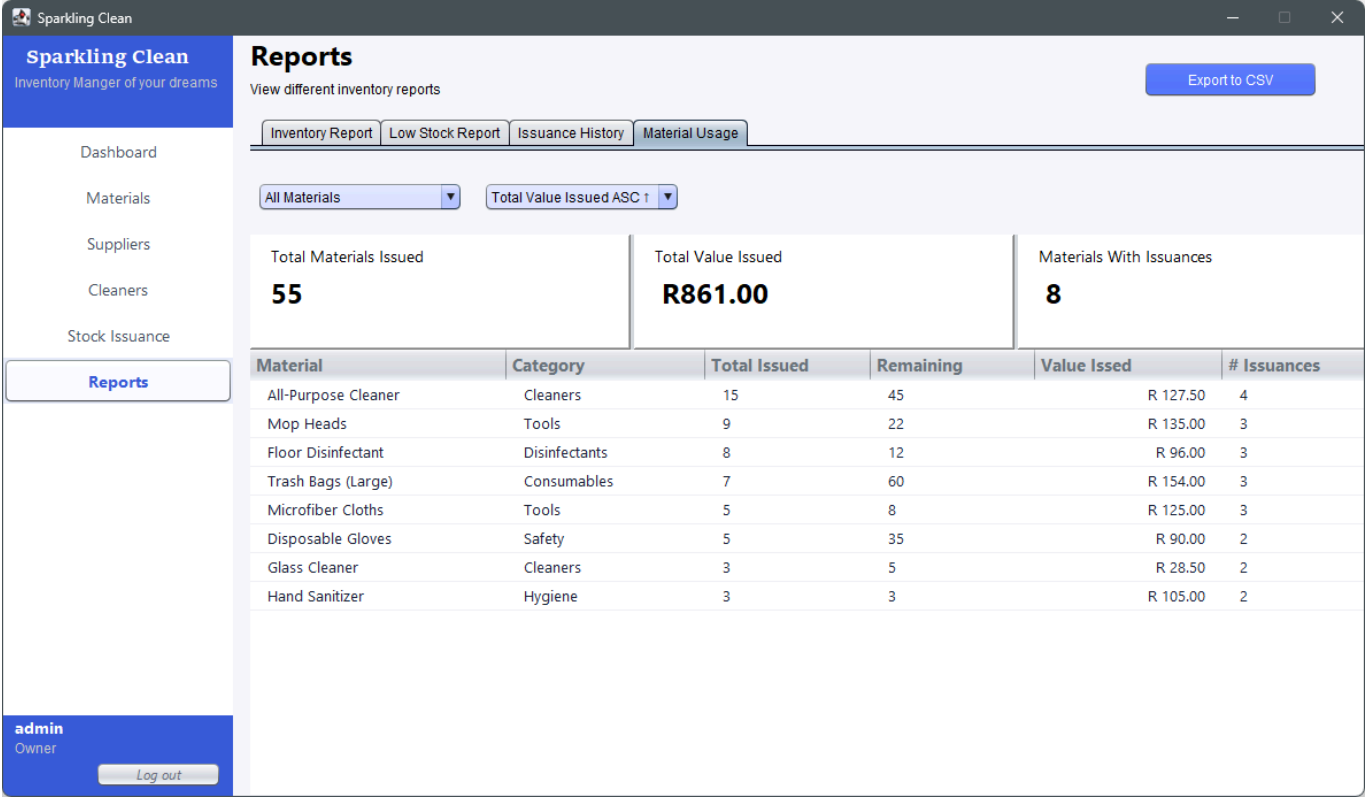
Inventory ReportLow Stock ReportIssuance HistoryMaterial Usage

All Cleaners

Showing22 recordsTotal Items Issued55

Date	Material	Cleaner	Quantity	Issued By	Notes
28 June 2026 09:15	All-Purpose Cleaner	Maria Santos	5	jdoe	Weekly supply for Science Building
27 June 2026 11:30	Microfiber Cloths	James Wilson	2	jdoe	Replace worn cloths
26 June 2026 14:00	Disposable Gloves	Patricia Garcia	3	jdoe	Routine replenishment
25 June 2026 10:45	Floor Disinfectant	Robert Johnson	4	jdoe	Deep cleaning request
24 June 2026 08:30	Trash Bags (Large)	Maria Santos	3	jdoe	Regular stock replenishment
23 June 2026 13:10	Glass Cleaner	James Wilson	2	jdoe	Library window cleaning
22 June 2026 15:20	Mop Heads	Patricia Garcia	4	jdoe	Administration floor upkeep
21 June 2026 09:05	Hand Sanitizer	Robert Johnson	2	jdoe	Hand sanitation stations replenishm...
20 June 2026 10:00	Mop Heads	Elena Rodriguez	2	jdoe	Mop head replacement for Science ...
19 June 2026 14:30	All-Purpose Cleaner	Sarah Jenkins	3	jdoe	Weekly supply restock
18 June 2026 08:45	Floor Disinfectant	Maria Santos	2	sjenkins	Floor disinfectant top-up
17 June 2026 11:15	Trash Bags (Large)	James Wilson	2	sjenkins	Trash bag restock for Library
16 June 2026 13:40	All-Purpose Cleaner	Patricia Garcia	4	jdoe	General cleaning supplies
15 June 2026 09:50	Microfiber Cloths	Robert Johnson	1	jdoe	Microfiber cloth replacement
14 June 2026 10:20	Disposable Gloves	Elena Rodriguez	2	jdoe	Gloves for lab cleanup

Material Report



Storekeeper

Starting screen: **A blank pannel, user must choose a option to navigate to**

Module	Access
Dashboard	Full view
Materials	Create, Read, Update, Delete
Suppliers	View only
Cleaners	Create, Read, Update, Delete
Stock Issuance	Create, Read, Update, Delete
Reports	Full access

The Storekeeper has the widest day-to-day operational access. Dashboard and Reports are identical to what the Owner sees; the difference shows up on Materials, Suppliers, Cleaners, and Stock Issuance, where the Storekeeper has full CRUD everywhere except Suppliers.

Dashboard

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

jdoe
Storekeeper

Log out

Dashboard

Overview of your cleaning supplies inventory

Inventory Value

R 3,159.00

Across 8 material types

Total Materials

8

of 190 units total

Low Stock Items

4

Needs attention

Active Cleaners

6

1 inactive

Recent Issuances

0

in the Last 7 days

Low stock items

Material	Current	Min	Supplier
Hand Sanitizer	3	10	EcoClean Solutions
Glass Cleaner	5	10	EcoClean Solutions
Microfiber Cloths	8	10	Janitorial Essentials
Floor Disinfectant	12	15	CleanPro Supplies

Recent Stock Issuances

Material	Issued To	Quantity	Date
All-Purpose Cleaner	Maria Santos	5	2026-06-28 09:15
Microfiber Cloths	James Wilson	2	2026-06-27 11:30
Disposable Gloves	Patricia Garcia	3	2026-06-26 14:00
Floor Disinfectant	Robert Johnson	4	2026-06-25 10:45
Trash Bags (Large)	Maria Santos	3	2026-06-24 08:30
Glass Cleaner	James Wilson	2	2026-06-23 13:10
Mop Heads	Patricia Garcia	4	2026-06-22 15:20

Materials (Full CRUD)

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

jdoe
Storekeeper

Log out

Materials

Manage your cleaning materials inventory

Search materials ...

Search

All Categories

Add new Material

Total Items

8

Low Stock

4

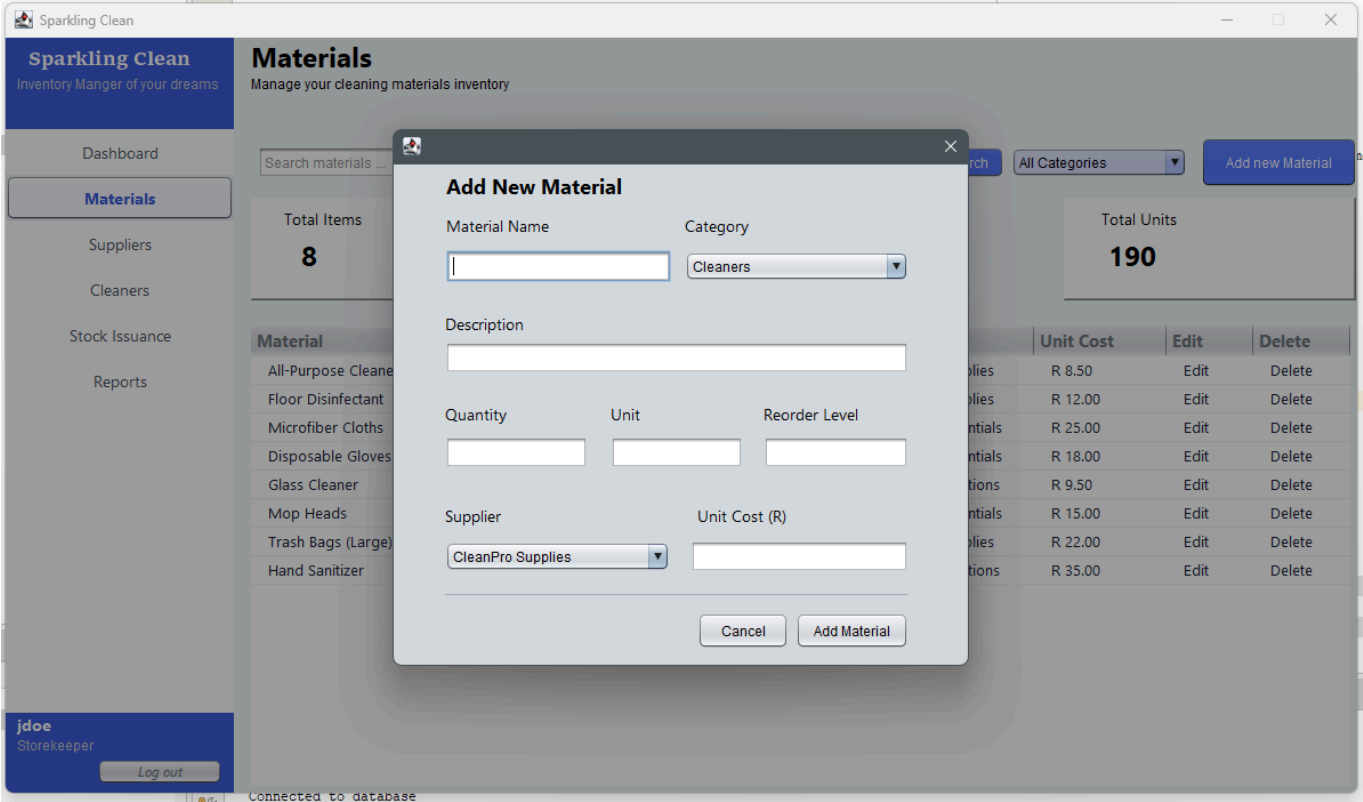
Total Units

190

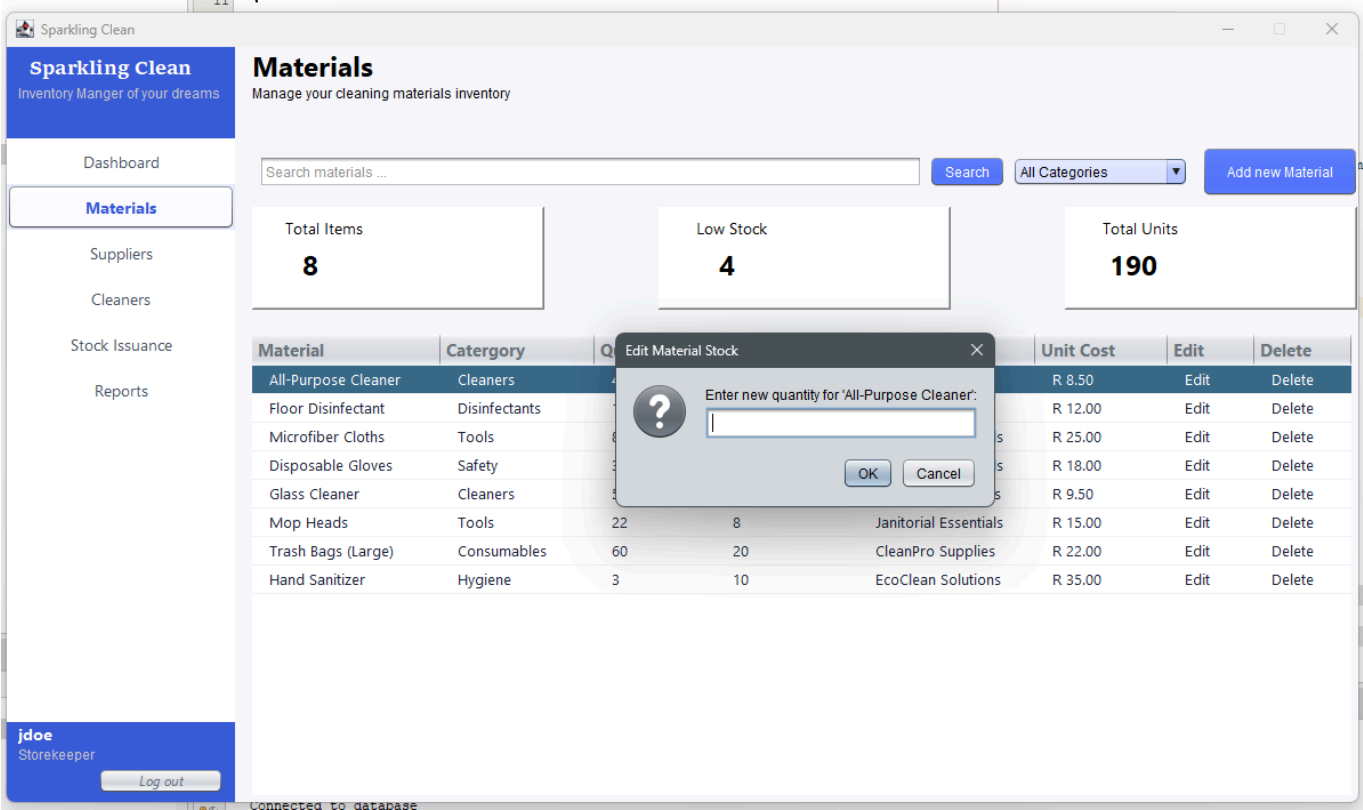
Material	Catergory	Quantity	Reorder Level	Supplier	Unit Cost	Edit	Delete
All-Purpose Cleaner	Cleaners	45	20	CleanPro Supplies	R 8.50	Edit	Delete
Floor Disinfectant	Disinfectants	12	15	CleanPro Supplies	R 12.00	Edit	Delete
Microfiber Cloths	Tools	8	10	Janitorial Essentials	R 25.00	Edit	Delete
Disposable Gloves	Safety	35	15	Janitorial Essentials	R 18.00	Edit	Delete
Glass Cleaner	Cleaners	5	10	EcoClean Solutions	R 9.50	Edit	Delete
Mop Heads	Tools	22	8	Janitorial Essentials	R 15.00	Edit	Delete
Trash Bags (Large)	Consumables	60	20	CleanPro Supplies	R 22.00	Edit	Delete
Hand Sanitizer	Hygiene	3	10	EcoClean Solutions	R 35.00	Edit	Delete

Material Pop-ups

Add Material (Create)



Edit Material (Update)



Delete Material (Delete)

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

jdoe

Storekeeper

Log out

Materials

Manage your cleaning materials inventory

Search materials ...

Search

All Categories

Add new Material

Total Items

8

Low Stock

4

Total Units

190

Material	Catergory	Unit Cost	Edit	Delete
All-Purpose Cleaner	Cleaners	R 8.50	Edit	Delete
Floor Disinfectant	Disinfectants	R 12.00	Edit	Delete
Microfiber Cloths	Tools	R 25.00	Edit	Delete
Disposable Gloves	Safety	R 18.00	Edit	Delete
Glass Cleaner	Cleaners	R 9.50	Edit	Delete
Mop Heads	Tools	R 15.00	Edit	Delete
Trash Bags (Large)	Consumables	R 22.00	Edit	Delete
Hand Sanitizer	Hygiene	R 35.00	Edit	Delete

Confirm Deletion

Are you sure you want to delete 'All-Purpose Cleaner'?

YesNo

Material	Category	Quantity	Low Stock	Supplier	Unit Cost	Edit	Delete
All-Purpose Cleaner	Cleaners	22	8	Janitorial Essentials	R 8.50	Edit	Delete
Floor Disinfectant	Disinfectants	60	20	CleanPro Supplies	R 12.00	Edit	Delete
Microfiber Cloths	Tools	3	10	EcoClean Solutions	R 25.00	Edit	Delete
Disposable Gloves	Safety				R 18.00	Edit	Delete
Glass Cleaner	Cleaners				R 9.50	Edit	Delete
Mop Heads	Tools				R 15.00	Edit	Delete
Trash Bags (Large)	Consumables				R 22.00	Edit	Delete
Hand Sanitizer	Hygiene				R 35.00	Edit	Delete

Connected to database

Suppliers (View Only)

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

jdoe

Storekeeper

Log out

Suppliers

Manage your cleaning supply vendors

Search suppliers by name, contact, or email ...

Search

Total Suppliers

3

Active

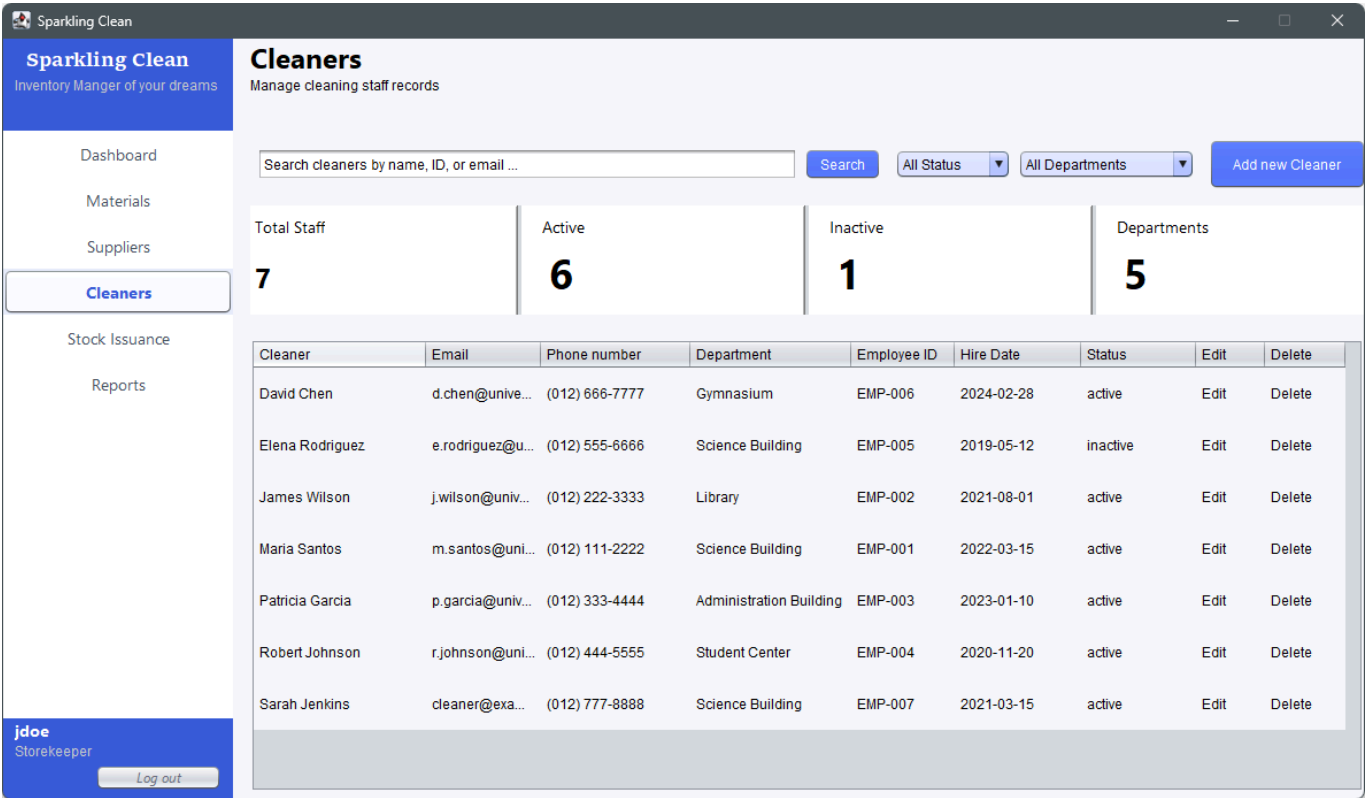
3

Total Materials Supplied

-

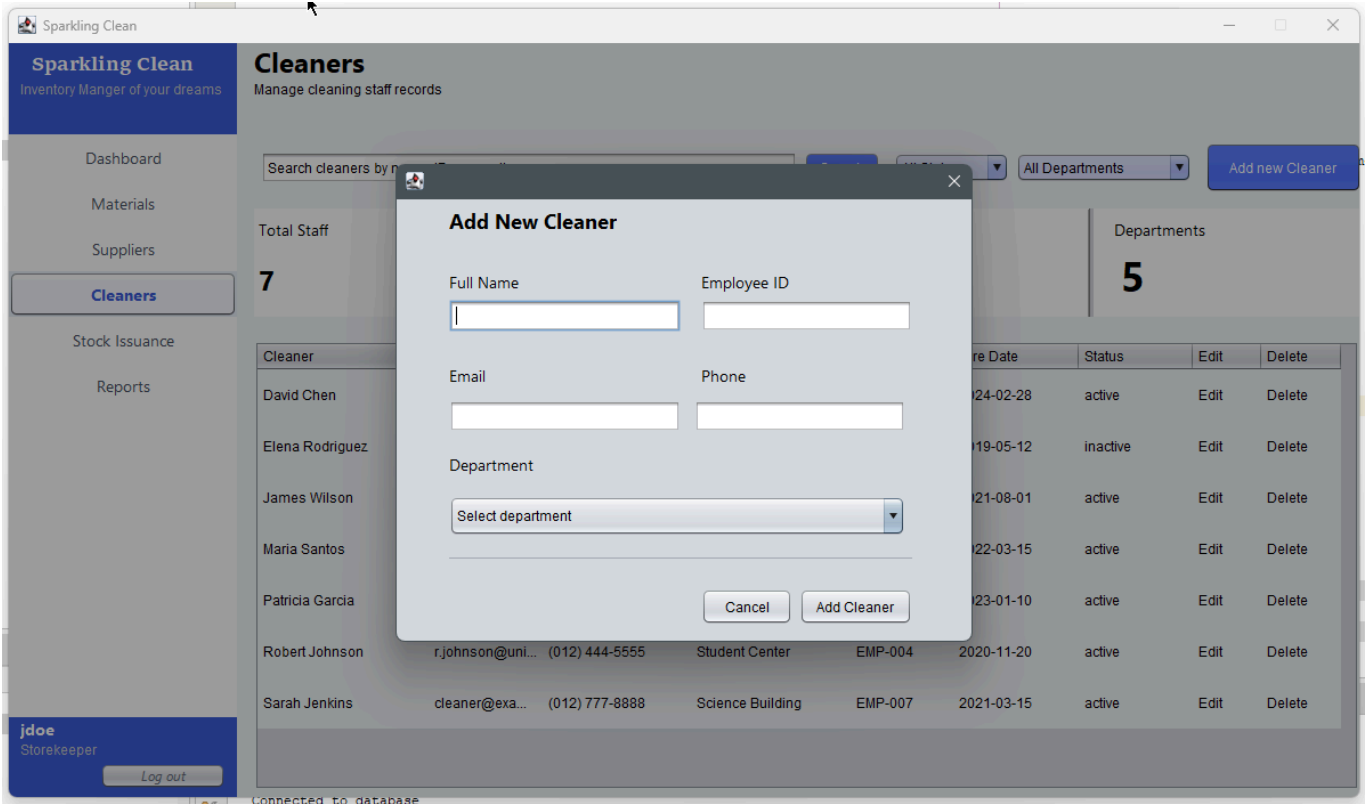
Supplier	Email	Phone number	Contact Person	Materials	Status
CleanPro Supplies	sales@cleanpro.com	(012) 019-8833	Alice Carter	-	-
EcoClean Solutions	support@ecocleansolutions.c...	(012) 017-7744	Diana Prince	-	-
Janitorial Essentials	orders@janitorialessentials.c...	(012) 014-9922	Bob Miller	-	-

Cleaners (Full CRUD)

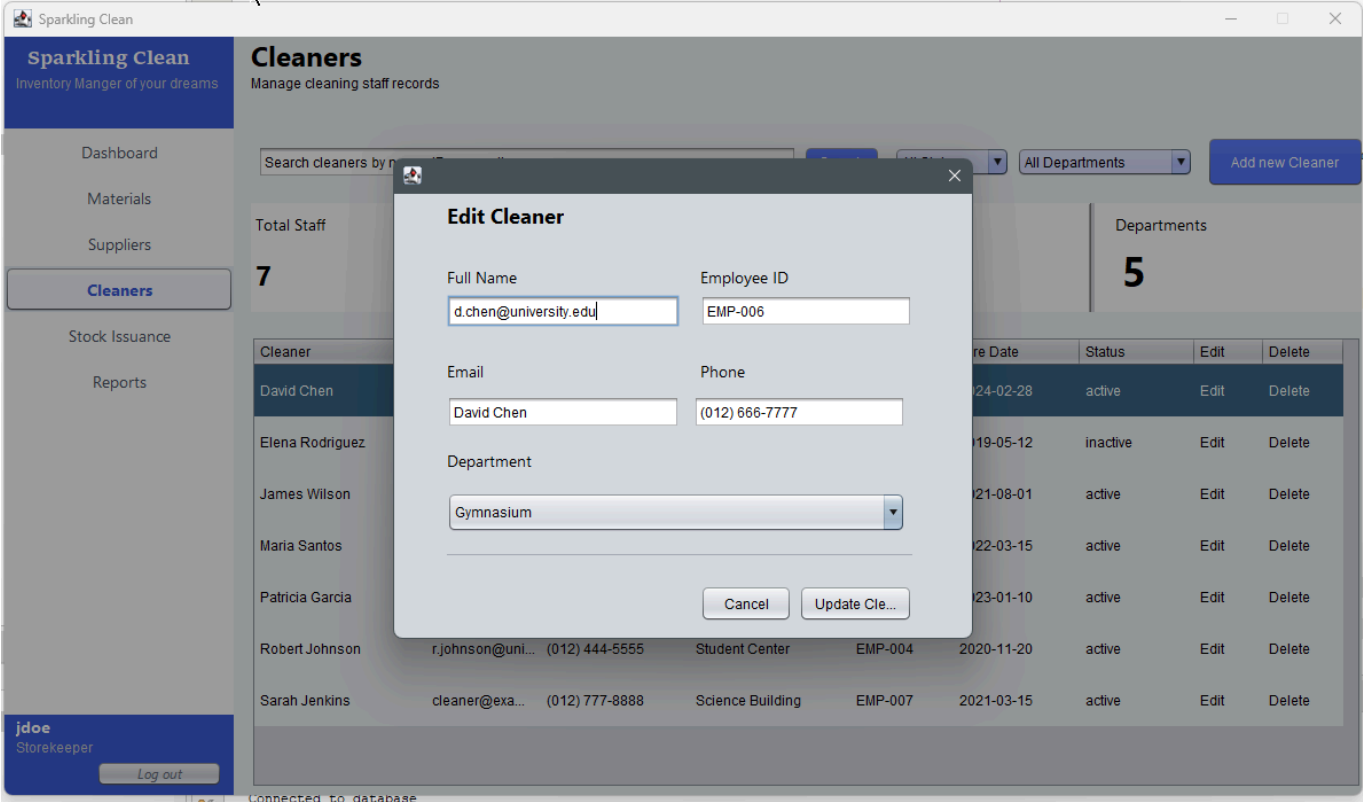


Cleaner Pop-ups

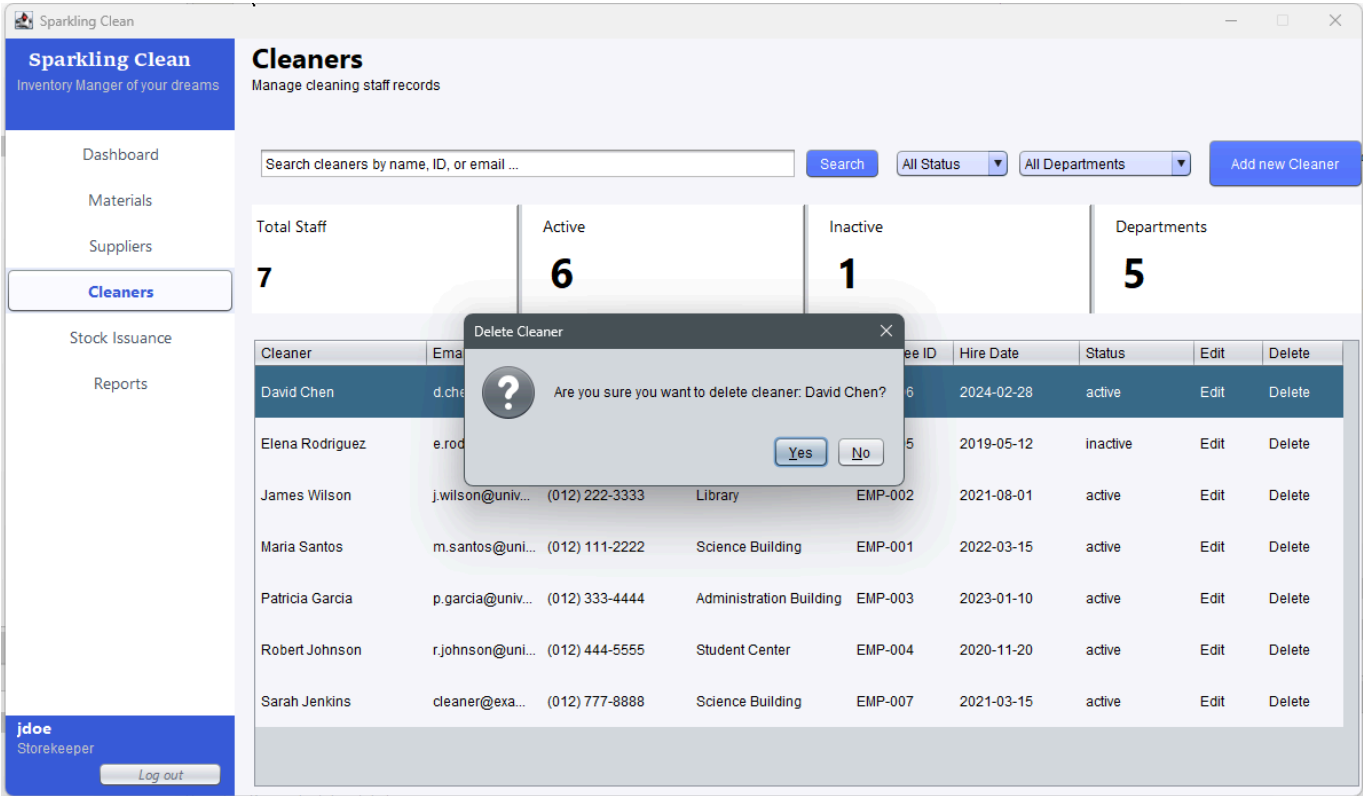
Add Cleaner (Create)



Edit Cleaner (Update)



Delete Cleaner (Delete)



Stock Issuance (Full CRUD)

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

jdoueStorekeeper

Log out

Stock Issuance

Check out the needed cleaning materials

Search by material ...

Search

Add New Issuance

Total Issuances

22

Total Items Issued

55

Low Stock items

4

Date	Material	Issued To	Quantity	Issued By	Notes	Edit	Delete
2026-06-28 09:15:0...	All-Purpose Cleaner	Maria San...	5	jdoue	Week...	Edit	Delete
2026-06-27 11:30:0...	Microfiber Cloths	James Wil...	2	jdoue	Repl...	Edit	Delete
2026-06-26 14:00:0...	Disposable Gloves	Patricia Ga...	3	jdoue	Routi...	Edit	Delete
2026-06-25 10:45:0...	Floor Disinfectant	Robert Jo...	4	jdoue	Deep...	Edit	Delete
2026-06-24 08:30:0...	Trash Bags (Large)	Maria San...	3	jdoue	Regu...	Edit	Delete
2026-06-23 13:10:0...	Glass Cleaner	James Wil...	2	jdoue	Libra...	Edit	Delete
2026-06-22 15:20:0...	Mop Heads	Patricia Ga...	4	jdoue	Admi...	Edit	Delete
2026-06-21 09:05:0...	Hand Sanitizer	Robert Jo...	2	jdoue	Hand...	Edit	Delete
2026-06-20 10:00:0...	Mop Heads	Elena Rod...	2	jdoue	Mop ...	Edit	Delete
2026-06-19 14:30:0...	All-Purpose Cleaner	Sarah Jenk...	3	jdoue	Week...	Edit	Delete
2026-06-18 08:45:0...	Floor Disinfectant	Maria San...	2	sjenkins	Floor...	Edit	Delete
2026-06-17 11:15:0...	Trash Bags (Large)	James Wil...	2	sjenkins	Trash...	Edit	Delete
2026-06-16 13:40:0...	All-Purpose Cleaner	Patricia Ga...	4	jdoue	Gene...	Edit	Delete
2026-06-15 09:50:0...	Microfiber Cloths	Robert Jo...	1	jdoue	Micr...	Edit	Delete
2026-06-14 10:20:0...	Disposable Gloves	Elena Rod...	2	jdoue	Glov...	Edit	Delete
2026-06-13 15:00:0...	Glass Cleaner	Sarah Jenk...	1	sjenkins	Glass...	Edit	Delete

Stock Issuance Pop-up

Issue Stock (Create)

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

jdoueStorekeeper

Log out

Stock Issuance

Check out the needed cleaning materials

Search by material ...

Search

Add New Issuance

Total Issuances

22

Low Stock items

4

Date	Material	Issued To	Quantity	Issued By	Notes	Edit	Delete
2026-06-28 09:15:0...	All-Purpose Cleaner	Maria San...	5	jdoue	Week...	Edit	Delete
2026-06-27 11:30:0...	Microfiber Cloths	James Wil...	2	jdoue	Repl...	Edit	Delete
2026-06-26 14:00:0...	Disposable Gloves	Patricia Ga...	3	jdoue	Routi...	Edit	Delete
2026-06-25 10:45:0...	Floor Disinfectant	Robert Jo...	4	jdoue	Deep...	Edit	Delete
2026-06-24 08:30:0...	Trash Bags (Large)	Maria San...	3	jdoue	Regu...	Edit	Delete
2026-06-23 13:10:0...	Glass Cleaner	James Wil...	2	jdoue	Libra...	Edit	Delete
2026-06-22 15:20:0...	Mop Heads	Patricia Ga...	4	jdoue	Admi...	Edit	Delete
2026-06-21 09:05:0...	Hand Sanitizer	Robert Jo...	2	jdoue	Hand...	Edit	Delete
2026-06-20 10:00:0...	Mop Heads	Elena Rod...	2	jdoue	Mop ...	Edit	Delete
2026-06-19 14:30:0...	All-Purpose Cleaner	Sarah Jenk...	3	jdoue	Week...	Edit	Delete
2026-06-18 08:45:0...	Floor Disinfectant	Maria San...	2	sjenkins	Floor...	Edit	Delete
2026-06-17 11:15:0...	Trash Bags (Large)	James Wil...	2	sjenkins	Trash...	Edit	Delete
2026-06-16 13:40:0...	All-Purpose Cleaner	Patricia Ga...	4	jdoue	Gene...	Edit	Delete
2026-06-15 09:50:0...	Microfiber Cloths	Robert Jo...	1	jdoue	Micr...	Edit	Delete
2026-06-14 10:20:0...	Disposable Gloves	Elena Rod...	2	jdoue	Glov...	Edit	Delete
2026-06-13 15:00:0...	Glass Cleaner	Sarah Jenk...	1	sjenkins	Glass...	Edit	Delete

New Stock Issuance

Select Material

Choose a material to issue

Select Cleaner

Choose a cleaner to issue to

Quantity

Date

Notes (Optional)

Cancel

Issue Stock

Edit Stock issuance (Update)

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

jdoe

Storekeeper

Log out

Stock Issuance

Check out the needed cleaning materials

Search by material ...

Search

Add New Issuance

Total Issuances

22

Total Items Issued

55

Low Stock items

4

Date	Material	Issued To	Quantity	Issued By	Notes	Edit	Delete
2026-06-28 09:15:0...	All-Purpose Cleaner				Week...	Edit	Delete
2026-06-27 11:30:0...	Microfiber Cloths				Repl...	Edit	Delete
2026-06-26 14:00:0...	Disposable Gloves				Routi...	Edit	Delete
2026-06-25 10:45:0...	Floor Disinfectant				Deep...	Edit	Delete
2026-06-24 08:30:0...	Trash Bags (Large)				Regu...	Edit	Delete
2026-06-23 13:10:0...	Glass Cleaner				Libra...	Edit	Delete
2026-06-22 15:20:0...	Mop Heads	Patricia Ga...	4	jdoe	Admi...	Edit	Delete
2026-06-21 09:05:0...	Hand Sanitizer	Robert Jo...	2	jdoe	Hand...	Edit	Delete
2026-06-20 10:00:0...	Mop Heads	Elena Rod...	2	jdoe	Mop ...	Edit	Delete
2026-06-19 14:30:0...	All-Purpose Cleaner	Sarah Jenk...	3	jdoe	Week...	Edit	Delete
2026-06-18 08:45:0...	Floor Disinfectant	Maria San...	2	sjenkins	Floor...	Edit	Delete
2026-06-17 11:15:0...	Trash Bags (Large)	James Wil...	2	sjenkins	Trash...	Edit	Delete
2026-06-16 13:40:0...	All-Purpose Cleaner	Patricia Ga...	4	jdoe	Gene...	Edit	Delete
2026-06-15 09:50:0...	Microfiber Cloths	Robert Jo...	1	jdoe	Micr...	Edit	Delete
2026-06-14 10:20:0...	Disposable Gloves	Elena Rod...	2	jdoe	Glov...	Edit	Delete
2026-06-13 15:00:0...	Glass Cleaner	Sarah Jenk...	1	sjenkins	Glass...	Edit	Delete

Delete stock issuance (Delete)

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

jdoe

Storekeeper

Log out

Stock Issuance

Check out the needed cleaning materials

Search by material ...

Search

Add New Issuance

Total Issuances

22

Total Items Issued

55

Low Stock items

4

Date	Material	Issued To	Quantity	Issued By	Notes	Edit	Delete
2026-06-28 09:15:0...	All-Purpose Cleaner				Week...	Edit	Delete
2026-06-27 11:30:0...	Microfiber Cloths				Repl...	Edit	Delete
2026-06-26 14:00:0...	Disposable Gloves				Routi...	Edit	Delete
2026-06-25 10:45:0...	Floor Disinfectant				Deep...	Edit	Delete
2026-06-24 08:30:0...	Trash Bags (Large)				Regu...	Edit	Delete
2026-06-23 13:10:0...	Glass Cleaner				Libra...	Edit	Delete
2026-06-22 15:20:0...	Mop Heads	Patricia Ga...	4	jdoe	Admi...	Edit	Delete
2026-06-21 09:05:0...	Hand Sanitizer	Robert Jo...	2	jdoe	Hand...	Edit	Delete
2026-06-20 10:00:0...	Mop Heads	Elena Rod...	2	jdoe	Mop ...	Edit	Delete
2026-06-19 14:30:0...	All-Purpose Cleaner	Sarah Jenk...	3	jdoe	Week...	Edit	Delete
2026-06-18 08:45:0...	Floor Disinfectant	Maria San...	2	sjenkins	Floor...	Edit	Delete
2026-06-17 11:15:0...	Trash Bags (Large)	James Wil...	2	sjenkins	Trash...	Edit	Delete
2026-06-16 13:40:0...	All-Purpose Cleaner	Patricia Ga...	4	jdoe	Gene...	Edit	Delete
2026-06-15 09:50:0...	Microfiber Cloths	Robert Jo...	1	jdoe	Micr...	Edit	Delete
2026-06-14 10:20:0...	Disposable Gloves	Elena Rod...	2	jdoe	Glov...	Edit	Delete
2026-06-13 15:00:0...	Glass Cleaner	Sarah Jenk...	1	sjenkins	Glass...	Edit	Delete

Reports (Full Access)

Inventory Report

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

jdoe

Storekeeper

Log out

Reports

View different inventory reports

Export to CSV

Inventory ReportLow Stock ReportIssuance HistoryMaterial Usage

Total Materials

8

Total Units

190

Total Value

R3,159.00

Low Stock Items

4

Search material or supplier

Search

Total Value ASC ↑

Material	Category	Quantity	Reorder Level	Supplier	Unit Cost	Total Value	Status
All-Purpose Cleaner	Cleaners	45	20	CleanPro Supplies	R 8.50	R 382.50	OK
Disposable Gloves	Safety	35	15	Janitorial Essentials	R 18.00	R 630.00	OK
Floor Disinfectant	Disinfectants	12	15	CleanPro Supplies	R 12.00	R 144.00	Low Stock
Glass Cleaner	Cleaners	5	10	EcoClean Solutions	R 9.50	R 47.50	Low Stock
Hand Sanitizer	Hygiene	3	10	EcoClean Solutions	R 35.00	R 105.00	Low Stock
Microfiber Cloths	Tools	8	10	Janitorial Essentials	R 25.00	R 200.00	Low Stock
Mop Heads	Tools	22	8	Janitorial Essentials	R 15.00	R 330.00	OK
Trash Bags (Large)	Consumables	60	20	CleanPro Supplies	R 22.00	R 1,320.00	OK

Low Stock Report

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

jdoe

Storekeeper

Log out

Reports

View different inventory reports

Export to CSV

Inventory ReportLow Stock ReportIssuance HistoryMaterial Usage

4 items below reorder level

Immediate attention required for restocking

Material	Category	Current	Reorder Level	Shortage	Supplier
Hand Sanitizer	Hygiene	3	10	7	EcoClean Solutions
Glass Cleaner	Cleaners	5	10	5	EcoClean Solutions
Floor Disinfectant	Disinfectants	12	15	3	CleanPro Supplies
Microfiber Cloths	Tools	8	10	2	Janitorial Essentials

Issuance Report

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

jdoe

Storekeeper

Log out

Reports

View different inventory reports

Export to CSV

Inventory ReportLow Stock ReportIssuance HistoryMaterial Usage

All Cleaners

Showing22 recordsTotal Items Issued55

Date	Material	Cleaner	Quantity	Issued By	Notes
28 June 2026 09:15	All-Purpose Cleaner	Maria Santos	5	jdoe	Weekly supply for Science Building
27 June 2026 11:30	Microfiber Cloths	James Wilson	2	jdoe	Replace worn cloths
26 June 2026 14:00	Disposable Gloves	Patricia Garcia	3	jdoe	Routine replenishment
25 June 2026 10:45	Floor Disinfectant	Robert Johnson	4	jdoe	Deep cleaning request
24 June 2026 08:30	Trash Bags (Large)	Maria Santos	3	jdoe	Regular stock replenishment
23 June 2026 13:10	Glass Cleaner	James Wilson	2	jdoe	Library window cleaning
22 June 2026 15:20	Mop Heads	Patricia Garcia	4	jdoe	Administration floor upkeep
21 June 2026 09:05	Hand Sanitizer	Robert Johnson	2	jdoe	Hand sanitation stations replenishm...
20 June 2026 10:00	Mop Heads	Elena Rodriguez	2	jdoe	Mop head replacement for Science ...
19 June 2026 14:30	All-Purpose Cleaner	Sarah Jenkins	3	jdoe	Weekly supply restock
18 June 2026 08:45	Floor Disinfectant	Maria Santos	2	sjenkins	Floor disinfectant top-up
17 June 2026 11:15	Trash Bags (Large)	James Wilson	2	sjenkins	Trash bag restock for Library
16 June 2026 13:40	All-Purpose Cleaner	Patricia Garcia	4	jdoe	General cleaning supplies
15 June 2026 09:50	Microfiber Cloths	Robert Johnson	1	jdoe	Microfiber cloth replacement
14 June 2026 10:20	Disposable Gloves	Elena Rodriguez	2	jdoe	Gloves for lab cleanup

Material Report

Sparkling Clean

Inventory Manger of your dreams

Dashboard

Materials

Suppliers

Cleaners

Stock Issuance

Reports

jdoe

Storekeeper

Log out

Reports

View different inventory reports

Export to CSV

Inventory ReportLow Stock ReportIssuance HistoryMaterial Usage

All MaterialsTotal Value Issued ASC ↑

Total Materials Issued55

Total Value IssuedR861.00

Materials With Issuances8

Material	Category	Total Issued	Remaining	Value Issed	# Issuances
All-Purpose Cleaner	Cleaners	15	45	R 127.50	4
Mop Heads	Tools	9	22	R 135.00	3
Floor Disinfectant	Disinfectants	8	12	R 96.00	3
Trash Bags (Large)	Consumables	7	60	R 154.00	3
Microfiber Cloths	Tools	5	8	R 125.00	3
Disposable Gloves	Safety	5	35	R 90.00	2
Glass Cleaner	Cleaners	3	5	R 28.50	2
Hand Sanitizer	Hygiene	3	3	R 105.00	2

Cleaner

Starting screen: A blank pannel, user must choose a option to navigate to

Module	Access
Materials	View only

Module	Access
Stock Issuance	Create, Read, Update, Delete, own entries only

The Cleaner's sidebar is limited to just Materials and Stock Issuance. The Dashboard, Suppliers, Cleaners, and Reports buttons don't appear at all for this role, since a Cleaner has no need for them. Within Stock Issuance, Cleaners can only see and manage entries tied to their own account.

Materials (View Only)

Sparkling Clean

Inventory Manger of your dreams

Materials

Stock Issuance

sjenkins

Cleaner

Log out

Materials

Manage your cleaning materials inventory

Search materials ...

Search

All Categories

Total Items

8

Low Stock

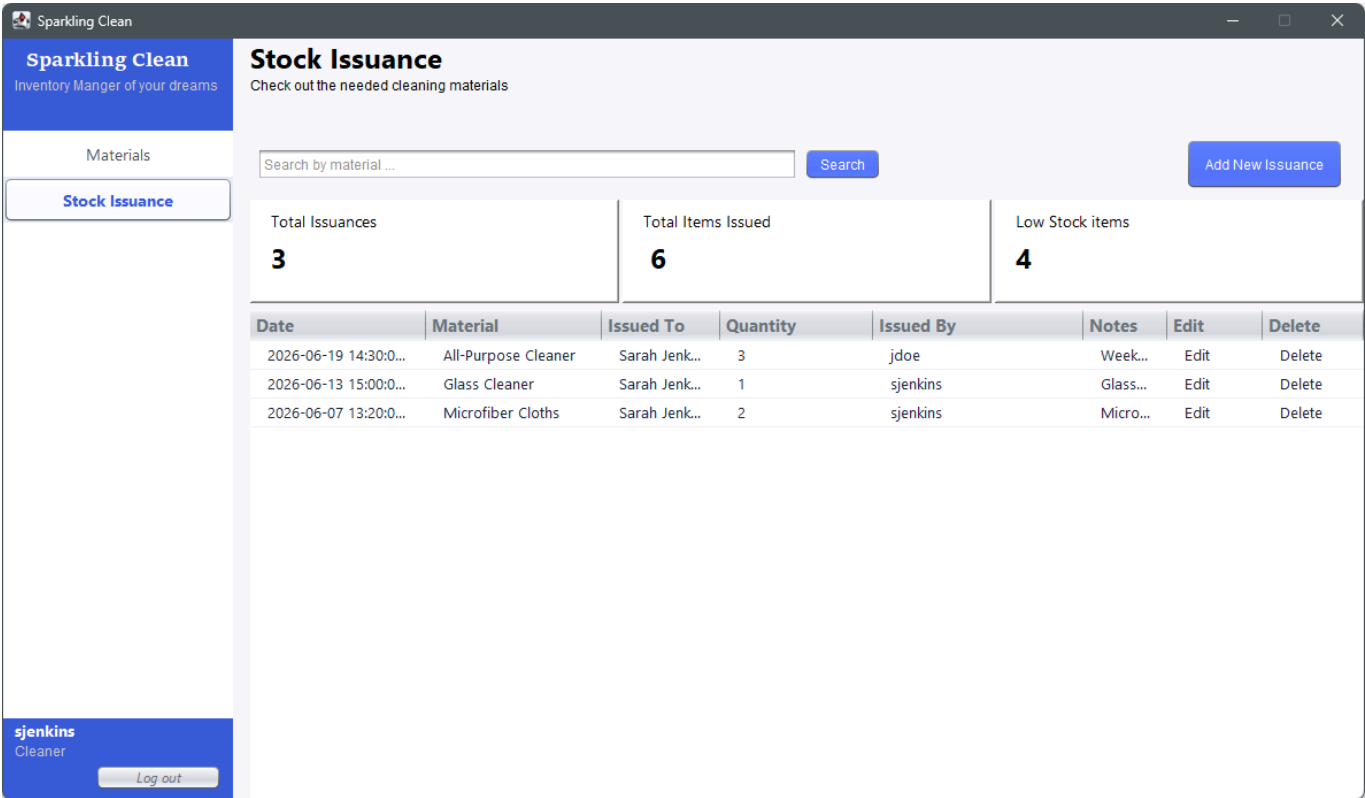
4

Total Units

190

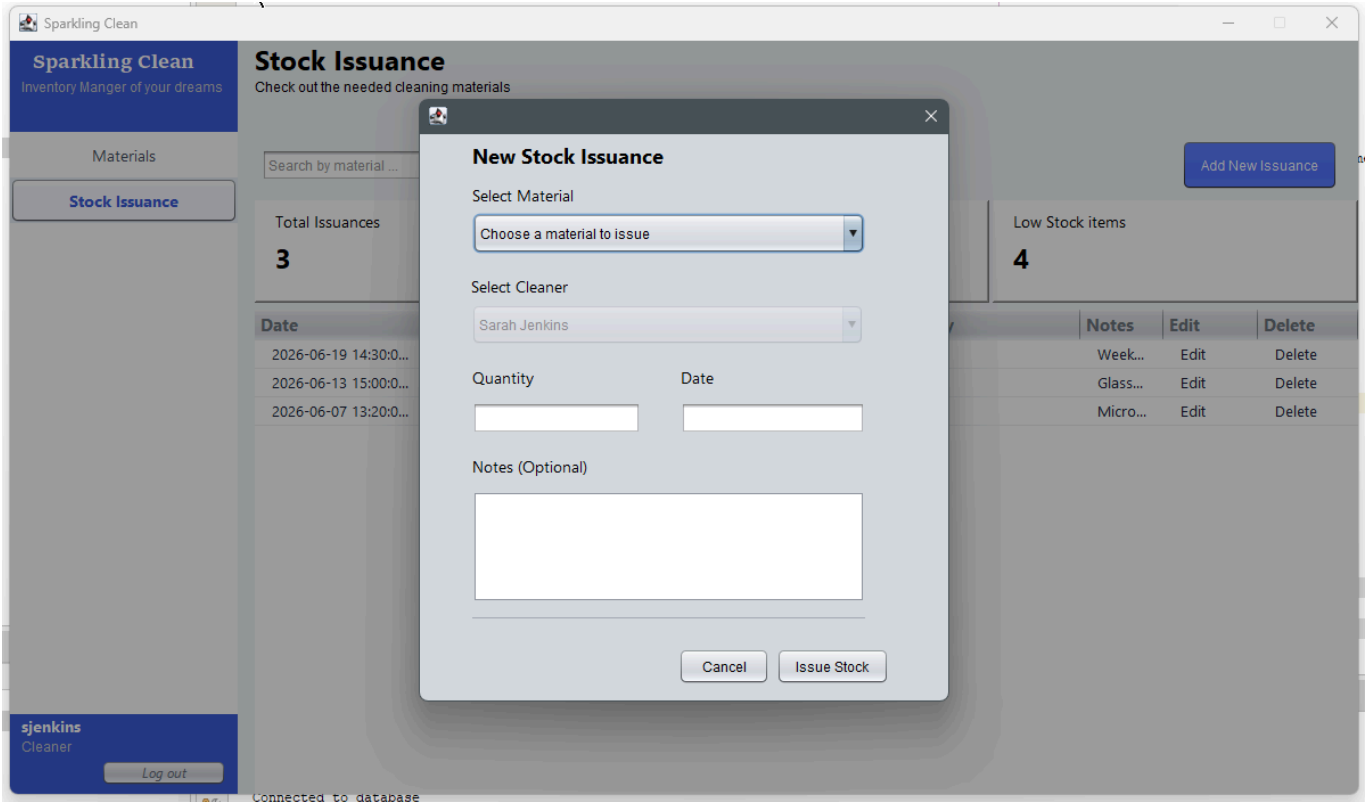
Material	Catergory	Quantity	Reorder Level	Supplier	Unit Cost
All-Purpose Cleaner	Cleaners	45	20	CleanPro Supplies	R 8.50
Floor Disinfectant	Disinfectants	12	15	CleanPro Supplies	R 12.00
Microfiber Cloths	Tools	8	10	Janitorial Essentials	R 25.00
Disposable Gloves	Safety	35	15	Janitorial Essentials	R 18.00
Glass Cleaner	Cleaners	5	10	EcoClean Solutions	R 9.50
Mop Heads	Tools	22	8	Janitorial Essentials	R 15.00
Trash Bags (Large)	Consumables	60	20	CleanPro Supplies	R 22.00
Hand Sanitizer	Hygiene	3	10	EcoClean Solutions	R 35.00

Stock Issuance (CRUD, Own Entries Only)

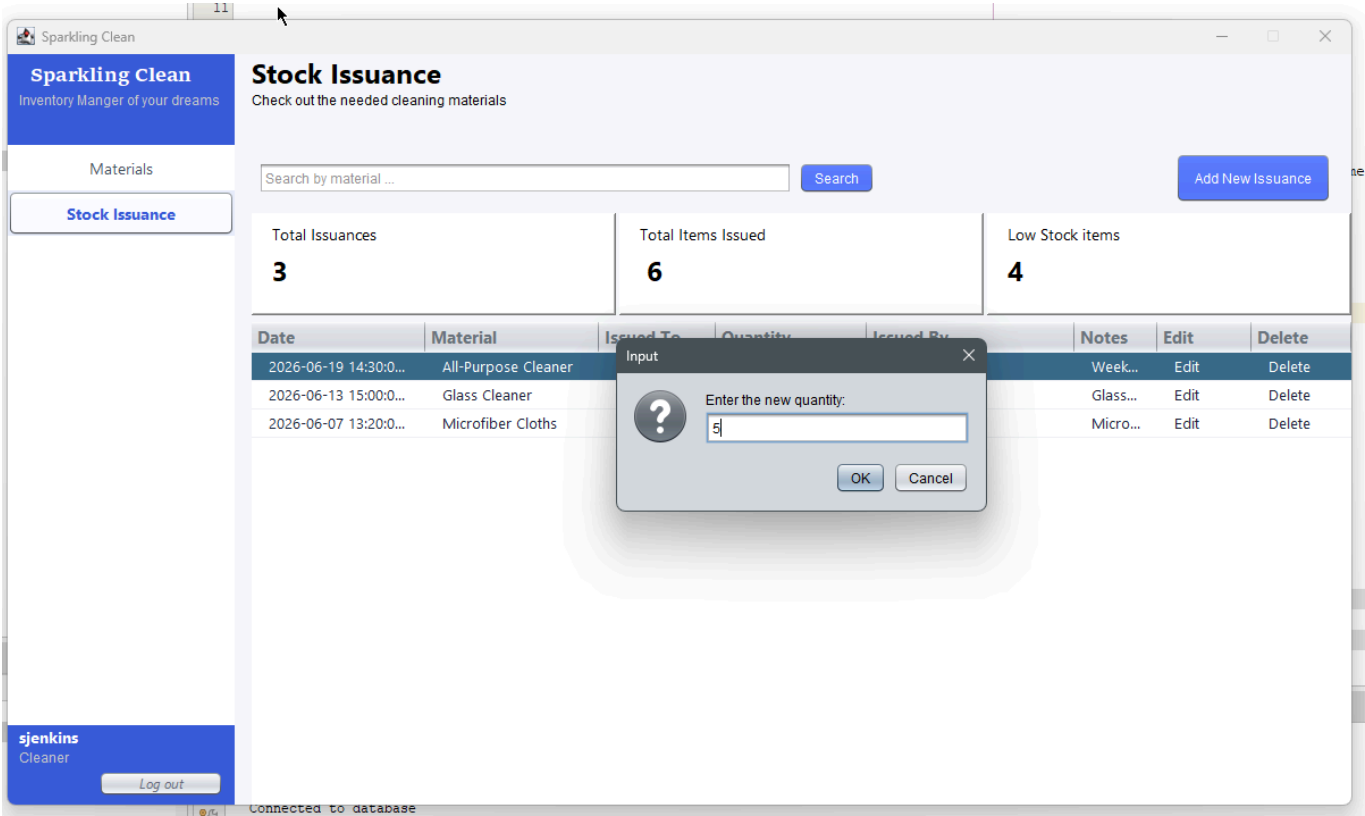


Stock Issuance Pop-up

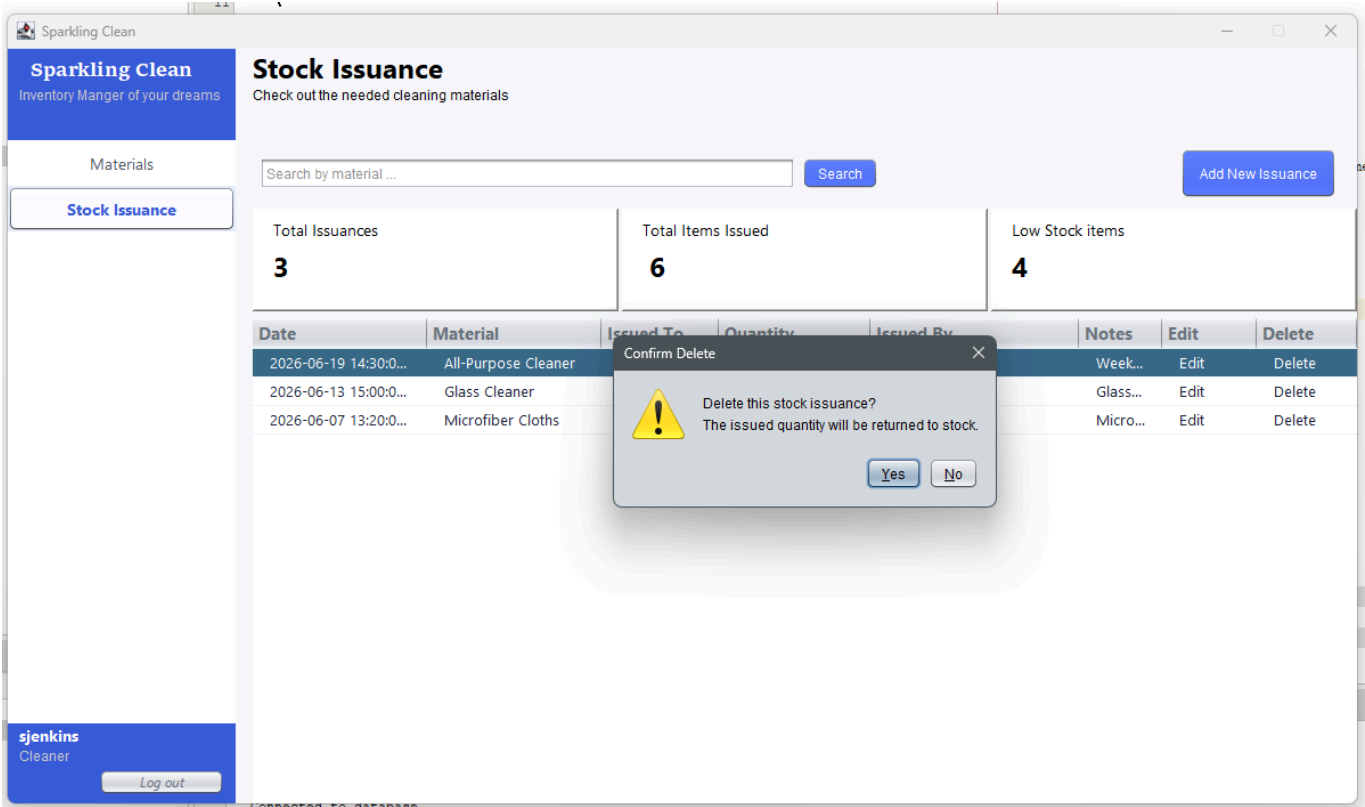
Issue Stock (Create)



Update Stock issuance (Update)

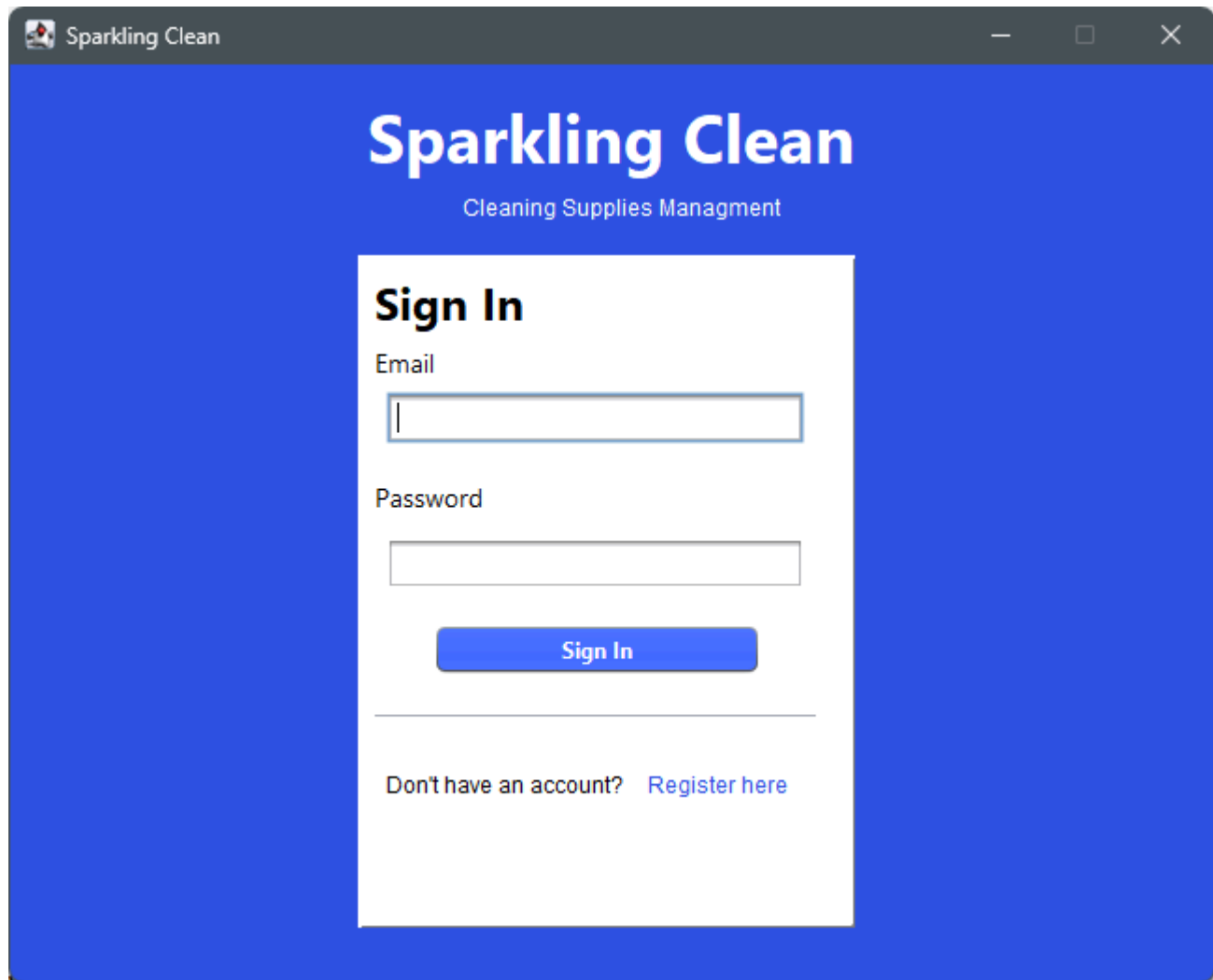


Delete stock issuance (Delete)



Login And Register Screenshots

Login Screen



The image shows a web browser window titled "Sparkling Clean" with a blue background. The main heading is "Sparkling Clean" in large white font, with the subtitle "Cleaning Supplies Management" below it. A white sign-in form is centered on the page. The form has a title "Sign In", an "Email" label above a text input field, a "Password" label above another text input field, and a blue "Sign In" button. Below the button is a horizontal line, and then the text "Don't have an account?" followed by a blue link "Register here".

Sparkling Clean

Cleaning Supplies Management

Sign In

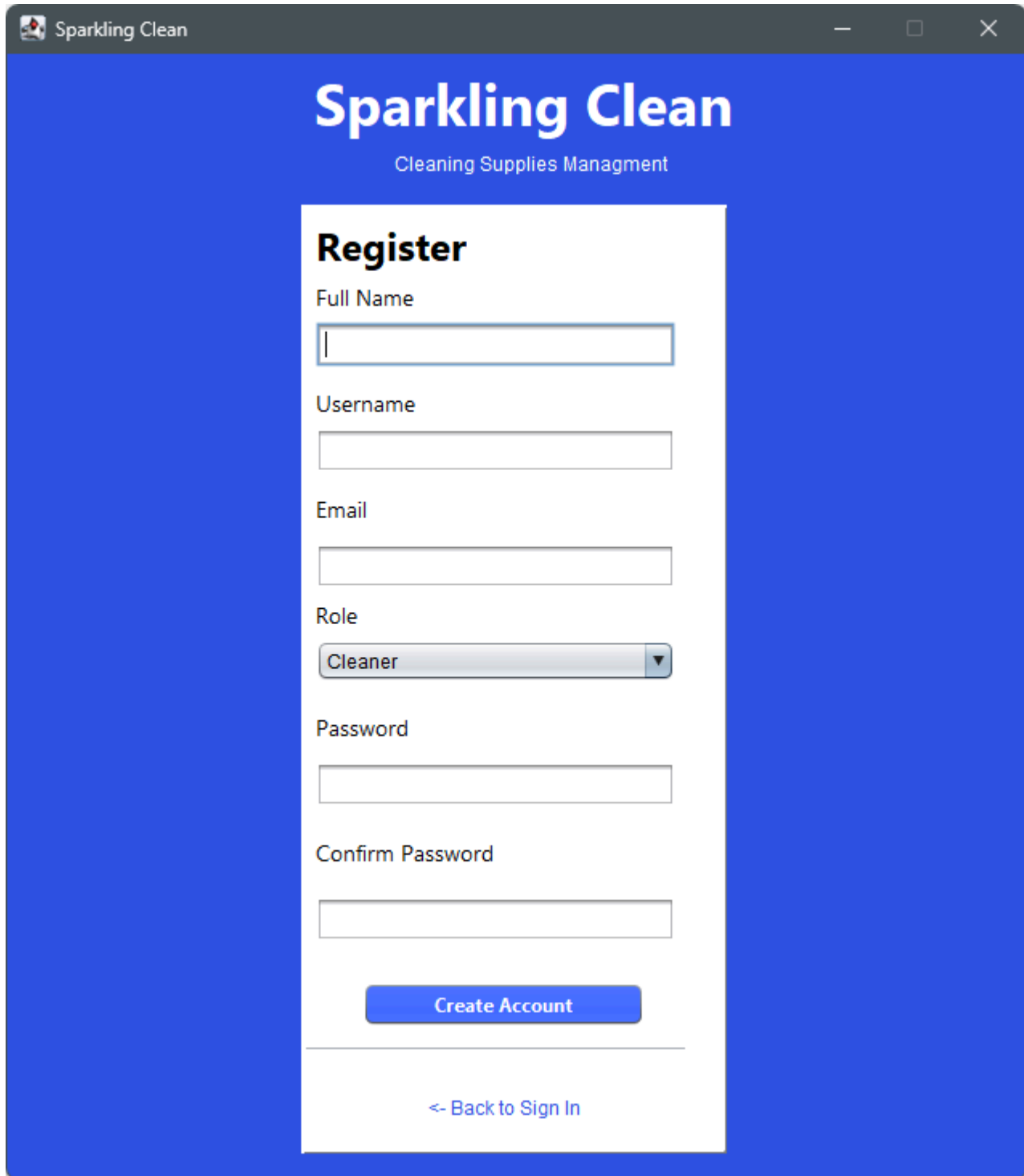
Email

Password

Sign In

Don't have an account? [Register here](#)

Register Screen



Sparkling Clean
Cleaning Supplies Management

Register

Full Name

Username

Email

Role

Password

Confirm Password

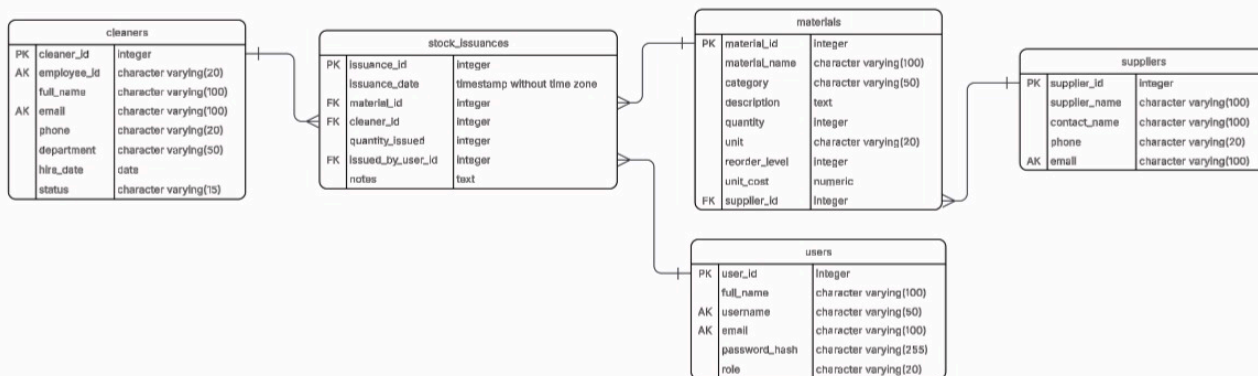
[Create Account](#)

[<- Back to Sign In](#)

Role-specific page screenshots (Dashboard, Materials, Suppliers, Cleaners, Stock Issuance, Reports) are shown under each role in the [Role-Based Access Control](#) section above, since each role sees a different starting screen and a different level of access on shared pages.

Database Schema (ERD)

The diagram below shows how the core tables (users, suppliers, materials, cleaners, stock_issuances) relate to one another.



Security

Passwords are never stored in plaintext. When a staff member registers or logs in, their password is hashed using SHA-256 and stored, via `src/utils/PasswordUtil.java`:

- `PasswordUtil.hash(password)`, hashes a plaintext password into its SHA-256 hex digest before it is stored (on registration) or compared (on login).
- `PasswordUtil.verify(password, storedHash)`, re-hashes an entered password and compares it against the stored hash, returning `true/false` without ever decrypting anything (SHA-256 is one-way).

```

public static String hash(String password) throws Exception {
    MessageDigest digest = MessageDigest.getInstance("SHA-256");
    byte[] encodedHash = digest.digest(password.getBytes(StandardCharsets.UTF_8));
    return bytesToHex(encodedHash);
}

public static boolean verify(String password, String storedHash) throws Exception
{
    return hash(password).equals(storedHash);
}
  
```

Note: this implementation hashes without a salt, so it's a simpler approach that is easier to implement.

The seeded login passwords shown elsewhere in this README (e.g. `Password@1234`) are the plaintext passwords you type into the login screen, the database only ever stores their SHA-256 hash.

Getting Started

Prerequisites

- **Windows**, this project is developed and tested on Windows; other operating systems are not officially supported
- **JDK 26**

- **Apache NetBeans 30** (the project uses NetBeans project files, so NetBeans is recommended over other IDEs)
- **PostgreSQL 18** (latest stable release), this project is configured for a local PostgreSQL setup only (no remote/cloud database support), so make sure PostgreSQL is installed and running on your own machine
- The **PostgreSQL JDBC driver** (`.jar`), see [Adding the PostgreSQL JDBC Driver](#) below

Installation

1. Go to the repository's GitHub page.
2. Click the green **Code** button, then **Download ZIP** (this downloads the `main` branch).
3. Extract the downloaded ZIP file to a folder on your computer.
4. Continue with the Database Setup and Opening the Project in NetBeans sections below.

Database Setup

This project uses a local PostgreSQL setup only, it does not support connecting to a remote or cloud-hosted database. You must install PostgreSQL on your own machine and create the database locally before running the app.

1. Install PostgreSQL if you haven't already, and make sure the PostgreSQL service is running.
2. Open pgAdmin and connect to your local PostgreSQL server.
3. In the left-hand tree, right-click on **Databases**, then select **Create** → **Database...**
4. In the dialog that opens, set the Database name to `sparkling_clean` (matching the name expected in `DBConnection.java`, see step 6 below), then click **Save**.
5. Run the project's SQL script against the new database to create the tables and load the mock/seed data. (Can be found at : PRG381_Project_PTA381_AM_Group-3\University Cleaning Inventory And Issuance System\database\schema_with_mockdata.sql)
 - In pgAdmin: right-click the `sparkling_clean` database → **Query Tool**, then open the `.sql` file (**File** → **Open...** within the Query Tool) and click **Execute/Run**.
 - Alternatively, from a terminal:

```
psql -U postgres -d sparkling_clean -f db/sparkling_clean.sql
```

This single script creates all required tables (users, suppliers, materials, cleaners, stock_issuances) and inserts the mock data used for testing (sample materials, suppliers, cleaners, and a default login for each role).

Once the script has run, you can log in with one of these seeded accounts to explore each role:

Role	Email	Password
Owner	<code>owner@example.com</code>	<code>Password@1234</code>

Role	Email	Password
Storekeeper	storekeeper@example.com	Password@1234
Cleaner	cleaner@example.com	Password@1234

6. Edit `src/utils/DBConnection.java` to match your local PostgreSQL setup. This is the file the app uses to connect to the database, so it must reflect the database name, username, and password you actually created:

```
package utils;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBConnection {
    // PostgreSQL JDBC driver class
    private static final String DRIVER = "org.postgresql.Driver";
    // JDBC URL to connect to the local PostgreSQL server + database
    private static final String JDBC_URL =
        "jdbc:postgresql://localhost:5432/sparkling_clean";
    // PostgreSQL local username and password required to access PostgreSQL
    private static final String USER = "postgres";
    private static final String PASSWORD = "password";

    public DBConnection() {
    }

    // Create a connection to the database
    public static Connection getConnection() throws ClassNotFoundException {
        Connection con = null;
        try {
            // Dynamically load the JDBC driver class at runtime
            Class.forName(DRIVER);
            con = DriverManager.getConnection(JDBC_URL, USER, PASSWORD);
            if (con != null) {
                System.out.println("Connected to database");
            }
        } catch (SQLException ex) {
            ex.printStackTrace();
        }
        return con;
    }
}
```

Update these three values to match what you configured in step 2:

Field	What to change it to
JDBC_URL	<code>jdbc:postgresql://localhost:<port>/<your_database_name></code> , keep <code>localhost</code> since this project only supports a local database; just update the port (if you changed it during install) and the database name if it differs from <code>sparkling_clean</code>
USER	The PostgreSQL username you connect with (default install often uses <code>postgres</code>)
PASSWORD	The password you set for that PostgreSQL user during installation

Without this step, the app will fail to connect and you'll see errors.

Opening the Project in NetBeans

1. Open Apache NetBeans.
2. Go to **File** → **Open Project...**
3. Browse to the folder where you extracted the ZIP file and select the project folder (the one containing `nbproject/` and `build.xml`).
4. Click **Open Project**.
5. Wait for NetBeans to index the project. If you see a red exclamation mark on the project name, see [Fixing JDK Version Errors](#) below.

Running the project

1. Make sure PostgreSQL is running and the database/tables/seed data are set up (see Database Setup above).
2. In NetBeans, right-click the project and select **Run**.
3. The `LoginFrame` should appear as the entry point, log in with one of the seeded accounts below (or register a new staff account) to continue:

Role	Email	Password
Owner	<code>owner@example.com</code>	<code>Password@1234</code>
Storekeeper	<code>storekeeper@example.com</code>	<code>Password@1234</code>
Cleaner	<code>cleaner@example.com</code>	<code>Password@1234</code>

Here is the consolidated section merged under **Resolving Errors**, cleanly organized into logical subsections:

Resolving Errors

1. Adding the PostgreSQL JDBC Driver (By default it should be there)

The project needs the PostgreSQL JDBC driver (`postgresql-<version>.jar`) on its classpath to connect to the database. This should already be added and should not be needed but if the problem does occur here is what to do

1. Download the driver `.jar` from jdbc.postgresql.org if you don't already have it.

2. In NetBeans, expand the project and right-click on **Libraries**.
 3. Select **Add JAR/Folder...**
 4. Browse to and select the downloaded `postgresql-<version>.jar` file.
 5. Click **Open**; the driver should now appear under Libraries in the project tree.
 6. Clean and build the project to confirm it compiles (right-click the project → **Clean and Build**).
-

2. Fixing JDK Version Errors in NetBeans

This project targets JDK 26. If NetBeans shows the project with a red marker or "missing JDK" / "invalid source/target release" type errors, it usually means the JDK configured on your machine doesn't match JDK 26.

What appears when the target JDK version or PostgreSQL JDBC driver is missing or misconfigured.

1. Right-click the project → **Properties**.
2. Go to **Sources**, and check the **Source/Binary Format** field, noting the Java version listed.
3. Go to **Libraries** → **Java Platform**, and check which JDK is selected.
4. If the required JDK isn't installed or listed:
 - Download and install the matching JDK version from [Adoptium](#) or [Oracle](#).
 - In NetBeans, go to **Tools** → **Java Platforms** → **Add Platform...** and point it to your new JDK installation folder.
5. Back in **Project Properties** → **Libraries**, select the correct Java Platform from the dropdown.
6. Apply changes, then **Clean and Build** the project again.

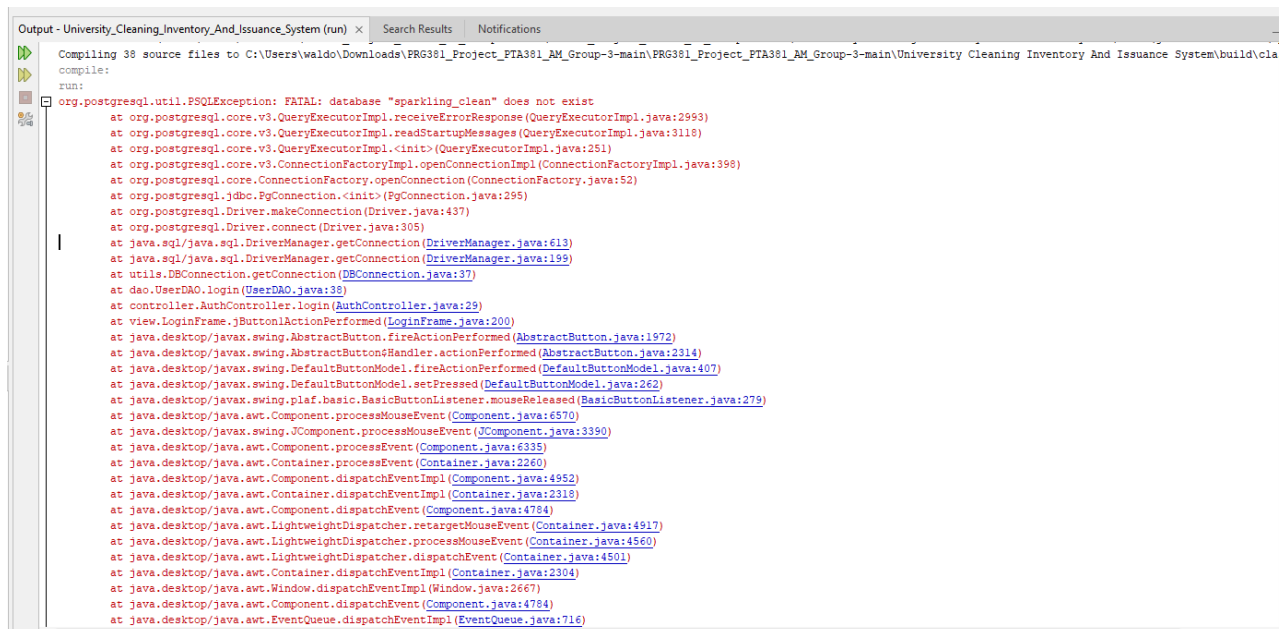
Reference view of what your Libraries / Java Platform configuration should look like once setup is correct.

3. Troubleshooting Common Setup & Runtime Issues

If something isn't working, check the reference screenshots below against what you are seeing—they cover the most common database setup and login authentication issues.

Database Setup Errors

- **Database Not Created in pgAdmin** *What you'll see if the `sparkling_clean` database was never created in pgAdmin.*

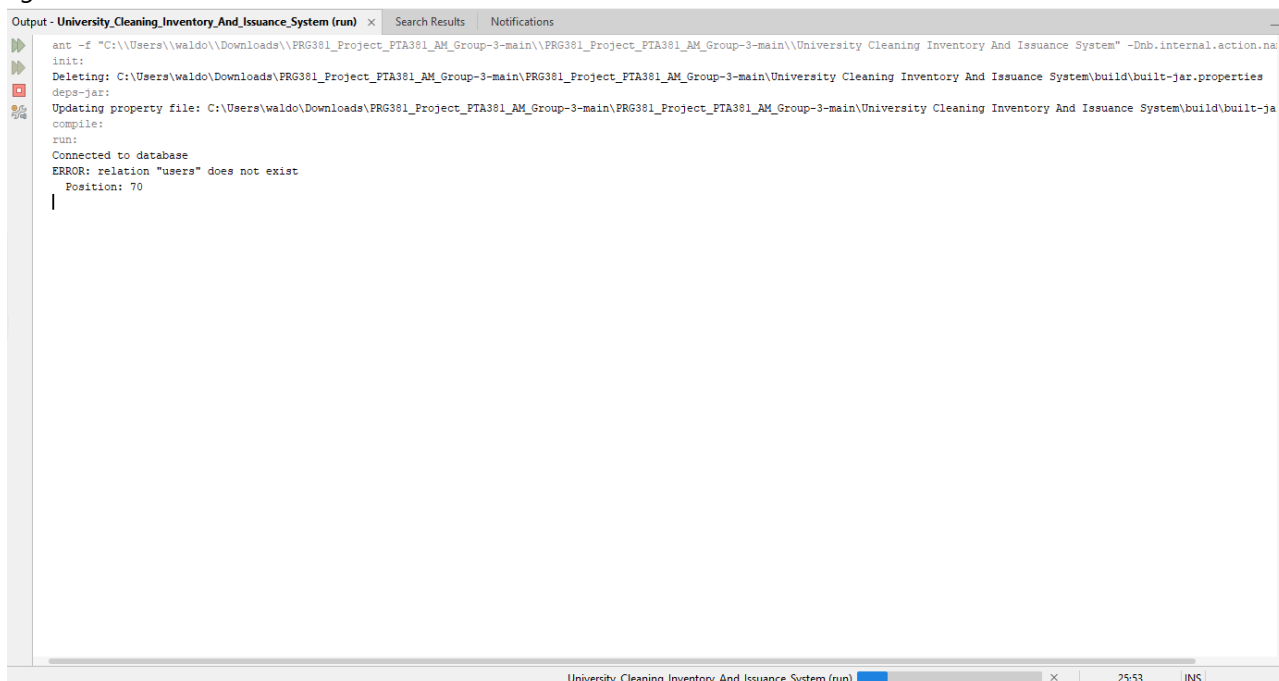


```

Output - University_Cleaning_Inventory_And_Issuance_System (run) x Search Results Notifications
Compiling 38 source files to C:\Users\waldo\Downloads\PRG381_Project_PTA381_AM_Group-3-main\PRG381_Project_PTA381_AM_Group-3-main\University_Cleaning_Inventory_And_Issuance_System\build\cla
compile:
run:
org.postgresql.util.PSQLException: FATAL: database "sparkling_clean" does not exist
    at org.postgresql.core.v3.QueryExecutorImpl.receiveErrorResponse(QueryExecutorImpl.java:2993)
    at org.postgresql.core.v3.QueryExecutorImpl.readStartupMessages(QueryExecutorImpl.java:3118)
    at org.postgresql.core.v3.QueryExecutorImpl.<init>(QueryExecutorImpl.java:251)
    at org.postgresql.core.v3.ConnectionFactoryImpl.openConnectionImpl(ConnectionFactoryImpl.java:398)
    at org.postgresql.core.ConnectionFactory.openConnection(ConnectionFactory.java:52)
    at org.postgresql.jdbc.PgConnection.<init>(PgConnection.java:295)
    at org.postgresql.Driver.makeConnection(Driver.java:437)
    at org.postgresql.Driver.connect(Driver.java:305)
    at java.sql/java.sql.DriverManager.getConnection(DriverManager.java:613)
    at java.sql/java.sql.DriverManager.getConnection(DriverManager.java:199)
    at utils.DBConnection.getConnection(DBConnection.java:37)
    at dao.UserDAO.login(UserDAO.java:38)
    at controller.AuthController.login(AuthController.java:29)
    at view.LoginFrame.jButtonActionPerformed(LoginFrame.java:200)
    at java.desktop/javawx.swing.AbstractButton.fireActionPerformed(AbstractButton.java:1972)
    at java.desktop/javawx.swing.AbstractButtonHandler.actionPerformed(AbstractButton.java:2314)
    at java.desktop/javawx.swing.DefaultButtonModel.fireActionPerformed(DefaultButtonModel.java:407)
    at java.desktop/javawx.swing.DefaultButtonModel.setPressed(DefaultButtonModel.java:262)
    at java.desktop/javawx.swing.plaf.basic.BasicButtonListener.mouseReleased(BasicButtonListener.java:279)
    at java.desktop/java.awt.Component.processMouseEvent(Component.java:6570)
    at java.desktop/javawx.swing.JComponent.processMouseEvent(JComponent.java:3390)
    at java.desktop/java.awt.Component.processEvent(Component.java:6335)
    at java.desktop/java.awt.Container.processEvent(Container.java:2260)
    at java.desktop/java.awt.Component.dispatchEventImpl(Component.java:4952)
    at java.desktop/java.awt.Container.dispatchEventImpl(Container.java:2318)
    at java.desktop/java.awt.Component.dispatchEvent(Component.java:4784)
    at java.desktop/java.awt.LightweightDispatcher.retargetMouseEvent(Container.java:4917)
    at java.desktop/java.awt.LightweightDispatcher.processMouseEvent(Container.java:4560)
    at java.desktop/java.awt.LightweightDispatcher.dispatchEvent(Container.java:4501)
    at java.desktop/java.awt.Container.dispatchEventImpl(Container.java:2304)
    at java.desktop/java.awt.Window.dispatchEventImpl(Window.java:2667)
    at java.desktop/java.awt.Component.dispatchEvent(Component.java:4784)
    at java.desktop/java.awt.EventQueue.dispatchEventImpl(EventQueue.java:716)

```

- **Database Not Populated** *What you'll see if the database exists but the SQL script was never executed against it.*



```

Output - University_Cleaning_Inventory_And_Issuance_System (run) x Search Results Notifications
ant -f "C:\Users\waldo\Downloads\PRG381_Project_PTA381_AM_Group-3-main\PRG381_Project_PTA381_AM_Group-3-main\University_Cleaning_Inventory_And_Issuance_System" -Dnb.internal.action.na
init:
Deleting: C:\Users\waldo\Downloads\PRG381_Project_PTA381_AM_Group-3-main\PRG381_Project_PTA381_AM_Group-3-main\University_Cleaning_Inventory_And_Issuance_System\build\build-jar.properties
deps-jar:
Updating property file: C:\Users\waldo\Downloads\PRG381_Project_PTA381_AM_Group-3-main\PRG381_Project_PTA381_AM_Group-3-main\University_Cleaning_Inventory_And_Issuance_System\build\build-jar
compile:
run:
Connected to database
ERROR: relation "users" does not exist
    Position: 70

```

- **DBConnection.java Not Updated** *What you'll see if src/utils/DBConnection.java still has default values instead of your own PostgreSQL credentials.*

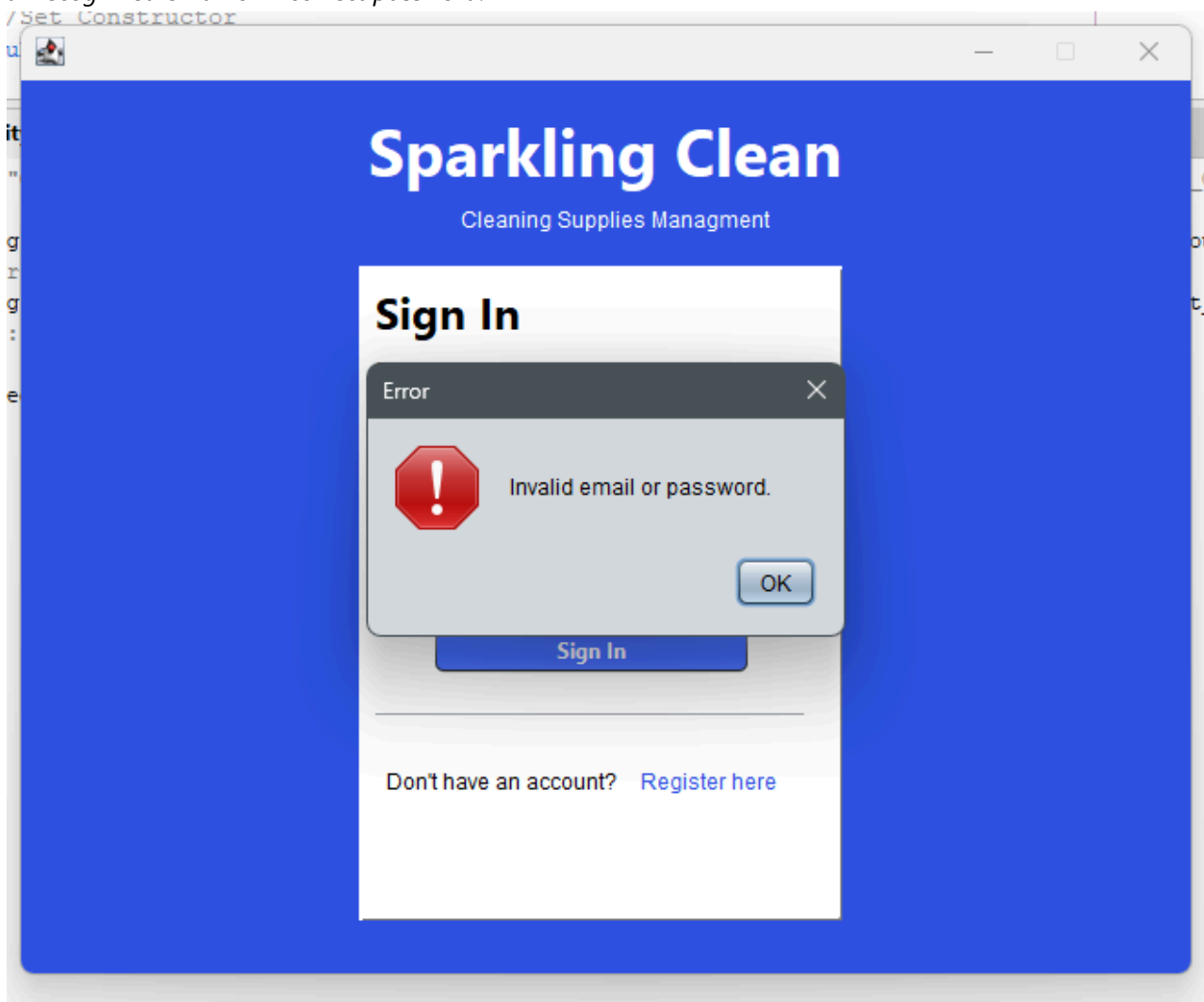
```

Output - University_Cleaning_Inventory_And_Issuance_System (run) x Search Results Notifications
compile:
run:
org.postgresql.util.PSQLException: FATAL: password authentication failed for user "postgres123"
at org.postgresql.core.v3.ConnectionFactoryImpl.doAuthentication(ConnectionFactoryImpl.java:837)
at org.postgresql.core.v3.ConnectionFactoryImpl.tryConnect(ConnectionFactoryImpl.java:309)
at org.postgresql.core.v3.ConnectionFactoryImpl.openConnectionImpl(ConnectionFactoryImpl.java:367)
at org.postgresql.core.ConnectionFactory.openConnection(ConnectionFactory.java:52)
at org.postgresql.jdbc.PgConnection.<init>(PgConnection.java:295)
at org.postgresql.Driver.makeConnection(Driver.java:437)
at org.postgresql.Driver.connect(Driver.java:305)
at java.sql/java.sql.DriverManager.getConnection(DriverManager.java:613)
at java.sql/java.sql.DriverManager.getConnection(DriverManager.java:199)
at utils.DBConnection.getConnection(DBConnection.java:37)
at dao.UserDAO.login(UserDAO.java:38)
at controller.AuthController.login(AuthController.java:29)
at view.LoginFrame.jButton1ActionPerformed(LoginFrame.java:200)
at java.desktop/javax.swing.AbstractButton.fireActionPerformed(AbstractButton.java:1972)
at java.desktop/javax.swing.AbstractButtonHandler.actionPerformed(AbstractButton.java:2314)
at java.desktop/javax.swing.DefaultButtonModel.fireActionPerformed(DefaultButtonModel.java:407)
at java.desktop/javax.swing.DefaultButtonModel.setPressed(DefaultButtonModel.java:262)
at java.desktop/javax.swing.plaf.basic.BasicButtonListener.mouseReleased(BasicButtonListener.java:279)
at java.desktop/java.awt.Component.processMouseEvent(Component.java:6570)
at java.desktop/javax.swing.JComponent.processMouseEvent(JComponent.java:3390)
at java.desktop/java.awt.Component.processEvent(Component.java:6335)
at java.desktop/java.awt.Container.processEvent(Container.java:2260)
at java.desktop/java.awt.Component.dispatchEventImpl(Component.java:4952)
at java.desktop/java.awt.Container.dispatchEventImpl(Container.java:2318)
at java.desktop/java.awt.Component.dispatchEvent(Component.java:4784)
at java.desktop/java.awt.LightweightDispatcher.retargetMouseEvent(Container.java:4917)
at java.desktop/java.awt.LightweightDispatcher.processMouseEvent(Container.java:4560)
at java.desktop/java.awt.LightweightDispatcher.dispatchEvent(Container.java:4501)
at java.desktop/java.awt.Container.dispatchEventImpl(Container.java:2304)
at java.desktop/java.awt.Window.dispatchEventImpl(Window.java:2667)
at java.desktop/java.awt.Component.dispatchEvent(Component.java:4784)
at java.desktop/java.awt.EventQueue.dispatchEventImpl(EventQueue.java:716)
at java.desktop/java.awt.EventQueue.dispatchEvent(EventQueue.java:693)
at java.desktop/java.awt.EventQueueDispatchThread.pumpOneEventForFilters(EventDispatchThread.java:203)
at java.desktop/java.awt.EventQueueDispatchThread.pumpEventsForFilter(EventDispatchThread.java:124)
at java.desktop/java.awt.EventQueueDispatchThread.pumpEventsForHierarchy(EventDispatchThread.java:113)

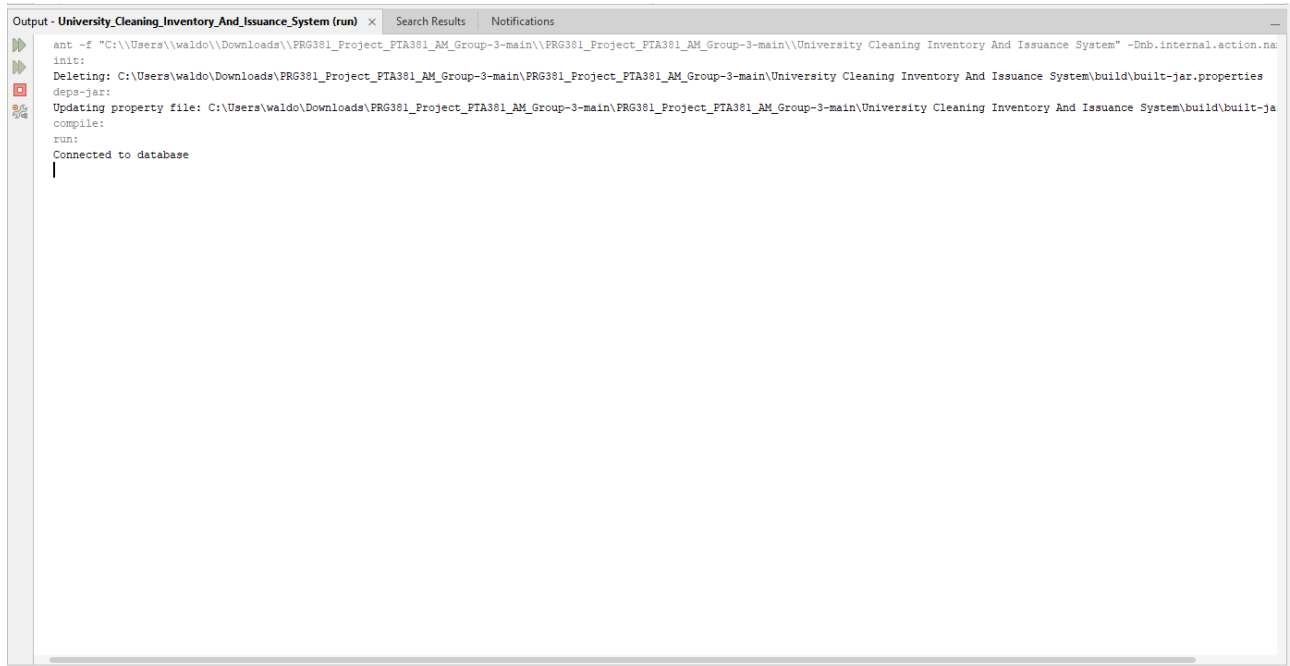
```

Authentication & Login Errors

- **Invalid Password or Email Pop up** The error prompt displayed when attempting to log in with an unrecognized email or incorrect password.



- **Failed Login Console Output** *The console output when the login fails but doesn't throw an unhandled exception, confirming active DB connection.*



```

Output - University_Cleaning_Inventory_And_Issuance_System (run) × Search Results Notifications
ant -f "C:\Users\waldo\Downloads\PRG381_Project_FTA381_AM_Group-3-main\PRG381_Project_FTA381_AM_Group-3-main\University_Cleaning_Inventory_And_Issuance_System" -Dnb.internal.action.name=init;
Deleting: C:\Users\waldo\Downloads\PRG381_Project_FTA381_AM_Group-3-main\PRG381_Project_FTA381_AM_Group-3-main\University_Cleaning_Inventory_And_Issuance_System\build\build-jar.properties
deps-jar:
Updating property file: C:\Users\waldo\Downloads\PRG381_Project_FTA381_AM_Group-3-main\PRG381_Project_FTA381_AM_Group-3-main\University_Cleaning_Inventory_And_Issuance_System\build\build-jar.properties
compile:
run:
Connected to database
|

```

Team, Group 3

Here is the updated contributors list with the corresponding `.java` files assigned to each person's functional contributions based on the project structure:

Contributors & Code Attribution

- **Waldo Blom** - 578068
- **Contribution:**
 - Created the initial UI shell & navigation for the entire application using NetBeans GUI Builder
 - Login and Register functionality (`LoginFrame.java`, `RegisterFrame.java`, `AuthController.java`, `UserDAO.java`)
 - User authentication and session management (`PasswordUtil.java`, `CurrentUser.java`, `User.java`)
 - Reports screen functionality implemented (`ReportsPnl.java`, `ReportsDAO.java`)
- **Tobie Jansen van Vuuren** - 602050
- **Contribution:**
 - Dashboard screen functionality (`DashboardPnl.java`)
 - Materials screen functionality (`MaterialsPnl.java`, `MaterialDAOInterface.java`, `Material.java`, `AddMaterialDialog.java`)
- **Arnold Mabope** - 603261

- **Contribution:**
- Suppliers screen functionality (`SuppliersPnl.java`, `SupplierController.java`, `SupplierService.java`, `SupplierDAO.java`, `Supplier.java`, `AddSuppliersDialog.java`, `EditSuppliersDialog.java`)
- Cleaners screen functionality (`CleanersPnl.java`, `CleanerController.java`, `CleanerService.java`, `CleanerDAO.java`, `Cleaner.java`, `AddCleanersDialog.java`, `EditCleanersDialog.java`)
- **Jonathan Rossouw** - 601761
- **Contribution:**
- Database creation and architecture design
- SQL schema creation script and mock data generation (`schema_with_mockdata.sql`, `ERD.jpeg`)
- Video presentation editing
- **Lindokuhle Shangase** - 602395
- **Contribution:**
- Stock Issuance screen functionality (`StockIssuancePnl.java`, `StockIssuanceController.java`, `StockIssuanceDAO.java`, `StockIssuance.java`, `AddStockIssuanceDialog.java`, `MaterialController.java`)